

Determinants of eHealth Systems Adoption: A Systematic Review and Multi-Case Analysis

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Abstract

Background: Despite sustained investment and policy attention, adoption of interoperable eHealth systems—including electronic health records (EHRs) and health information exchanges (HIEs)—remains inconsistent across organizations and national contexts. Existing frameworks have examined isolated factors without integrating them into a unified, operationalizable model.

Objective: This study aims to identify and synthesize determinants of eHealth systems adoption across diverse implementation contexts and develop a consolidated model applicable to healthcare leaders and system architects.

Methods: An integrative evidence synthesis combining a PRISMA 2020-guided systematic review with a structured comparative multi-case analysis. Database searches of PubMed/MEDLINE and IEEE Xplore, executed in June 2026, identified 1,688 records; Scopus, Web of Science, and CINAHL could not be searched owing to unavailable institutional subscription access. After deduplication, 1,680 records were screened against a priori criteria in two documented stages, 549 reports were assessed for eligibility, and 501 studies were included for evidence extraction. The multi-case analysis examined four national implementation contexts: U.S. ONC HIE programs, NHS Shared Care Records (UK), Estonia’s national eHealth system, and the U.S. federal EHR modernization, with the Department of Veterans Affairs and Department of Defense programs analysed separately. Findings were integrated using the U-T-I-O Model, which is additionally operationalized into a bounded 0–1 alignment index (UAI).

Results: Adoption determinants clustered across four interdependent domains: User Readiness (usability, workflow fit, training, perceived usefulness), Technical Interoperability (system quality, semantic/syntactic interoperability, data completeness),

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Institutional Alignment (regulatory mandates, national digital health strategy, information governance), and Organizational Execution (leadership, change management, stakeholder engagement, resource capacity). Technical interoperability is necessary but insufficient for sustained adoption. Estonia showed the highest alignment across all four domains; the VA/DoD case demonstrated how strong institutional mandate can be undermined by gaps in user readiness and organizational execution.

Conclusions: The U-T-I-O Model provides an operationalizable framework for assessing eHealth adoption readiness and diagnosing domain-specific implementation gaps. Sustainable adoption requires coordinated alignment across all four domains. The model should be interpreted as an implementation-readiness and adoption-maturity framework, not as a validated predictor of patient-level clinical outcomes; establishing clinical outcome validity through the proposed U-T-I-O-OM extension is the primary direction for future research.

Keywords: eHealth adoption; electronic health records; health information exchange; interoperability; systematic review; U-T-I-O Model; implementation science

Summary Table

What this systematic review adds

- eHealth adoption is determined by alignment across four interacting domains—User Readiness, Technical Interoperability, Institutional Alignment, and Organizational Execution—not by any single domain in isolation.
- Across 501 included studies and four national implementation cases, most studies presented determinants spanning three or more domains and none identified a single-domain explanation as sufficient; multi-domain convergence emerged as the decisive predictor of adoption maturity.
- Strong institutional mandate and substantial technical investment are necessary but insufficient: the VA/DoD case demonstrates how realised gaps in user readiness, technical interoperability, and organizational execution constrain adoption value despite congressional mandate and sustained investment.
- The U-T-I-O Model functions as both a descriptive framework and a diagnostic tool: it is operationalized into a bounded 0–1 alignment index (UAI) that scores an implementation’s readiness and localizes domain-level misalignment risks before deployment, illustrated across the national cases (including the U.S. VA and DoD programs scored separately).

1 Introduction

Healthcare systems worldwide are undergoing a fundamental digital transformation. The adoption of interoperable electronic health record (EHR) systems, health information exchanges (HIEs), and broader eHealth infrastructure has become a central priority for governments, healthcare organizations, and technology vendors alike. In the United States, the Health Information Technology for Economic and Clinical Health (HITECH) Act of 2009 catalyzed a national investment exceeding \$35 billion in EHR adoption incentives [1, 2]. Internationally, national digital health strategies in the United Kingdom, Estonia, Australia, Canada, and across the European Union have similarly driven large-scale investments in interoperable health information infrastructure.

Despite this sustained policy attention and financial investment, eHealth adoption outcomes remain strikingly uneven. Some health systems—such as Estonia’s national eHealth platform—have achieved near-comprehensive coverage, high clinician engagement, and measurable improvements in care coordination [3, 4]. Others, including some of the most heavily funded programs in the United States, such as the Department of Veterans Affairs (VA) and Department of Defense (DoD) joint EHR modernization initiative, continue to face significant usability challenges, implementation delays, and user dissatisfaction, even years after deployment [5, 6]. This paradox—substantial investment yielding inconsistent outcomes—motivates the central inquiry of this study.

The adoption science literature offers several frameworks for understanding why technology implementations succeed or fail. The DeLone and McLean Information Systems (IS) Success Model [7] identifies system quality, information quality, use, and net benefits as critical success dimensions. The Technology-Organization-Environment (TOE) Framework [8] situates adoption decisions within the interacting contexts of technological readiness, organizational capacity, and environmental pressures. While individually illuminating, these frameworks were not developed with the specific complexities of healthcare in mind—a domain characterized by high regulatory density, sensitive data governance requirements, deeply embedded clinical workflows, and multi-stakeholder governance structures. Prior systematic reviews of eHealth adoption have tended to examine isolated determinants—usability, interoperability standards, policy mandates, or organizational leadership—without integrating them into a unified, operationalizable model applicable across diverse implementation contexts [9–11].

This study addresses that gap and extends a sustained line of inquiry. The systematic review reported here builds directly on the author’s prior qualitative research [12], which identified adoption strategies from interviews with senior IT leaders in healthcare organizations; the current study extends that practitioner-grounded foundation by synthesizing the broader systematic-review literature and validating the resulting framework against four

publicly documented national implementations, consolidating the prior qualitative findings into a formalized analytical model. This article reports a PRISMA-guided systematic review of the peer-reviewed literature on eHealth adoption determinants, synthesized alongside a comparative multi-case analysis of four major national eHealth implementations documented through primary and grey-literature sources. Drawing on both the systematic review evidence and the case analyses, I develop and validate the **U-T-I-O Model** of eHealth adoption—a structured framework that integrates **U**ser Readiness, **T**echnical Interoperability, **I**nstitutional Alignment, and **O**rganizational Execution as the four interdependent determinants of successful eHealth systems adoption.

The U-T-I-O Model advances the field by synthesizing determinants from empirical research and real-world evidence into a single operationalizable construct that explicitly incorporates the institutional and policy dimensions distinguishing healthcare from other IS adoption contexts. It provides leaders, system architects, and policymakers with a practical diagnostic tool for assessing adoption readiness and identifying domain-specific gaps. The scope of this study is explicitly bounded to *implementation readiness and maturity*: the model characterizes the organizational, technical, institutional, and human conditions under which eHealth systems are positioned for sustainable adoption. It does not extend to direct measurement of patient clinical outcomes; the relationship between implementation maturity and clinical benefit is identified as the primary direction for future research and is addressed in the Conclusion.

This study contributes to eHealth adoption research in three ways. First, it consolidates fragmented adoption determinants into a four-domain model specific to interoperable healthcare systems. Second, it validates the model against four publicly documented national implementation contexts (with the U.S. VA and DoD programs analysed separately) and operationalizes it into a bounded alignment index. Third, it distinguishes implementation-readiness validity from clinical outcome validity, establishing a pathway for future U-T-I-O-OM validation.

This article is organized as follows: Section 2 reviews the theoretical foundations; Section 3 presents the research questions; Section 4 describes the methodology; Section 5 reports findings; Section 6 discusses implications; and Sections 7–8 address limitations and conclusions.

2 Background and Theoretical Foundation

2.1 DeLone and McLean Information Systems Success Model

The DeLone and McLean IS Success Model is among the most widely applied frameworks in information systems research [7]. Updated in 2003, the model identifies six dimensions of IS success—system quality, information quality, service quality, system use, user satisfaction, and net benefits—operating as a causal chain in which technical quality shapes use and satisfaction, which in turn determine realized benefits. In healthcare, the model has been applied to evaluate EHR implementations [13] and identify sources of user resistance to clinical information systems.

2.2 Technology-Organization-Environment Framework

The Technology-Organization-Environment (TOE) framework [8] situates adoption decisions within three contextual domains: the *technological* context (available and emerging technologies), the *organizational* context (structure, resources, and human capital), and the *environmental* context (regulatory pressures, competitive forces, and inter-organizational relationships). In healthcare, the environmental context is especially determinative: organizations adopt health IT primarily in response to regulatory requirements (HITECH, 21st Century Cures Act), quality reporting mandates, and patient safety imperatives rather than competitive market signals [14, 15]. The TOE framework’s explicit attention to this regulatory environment makes it particularly well-suited to the healthcare domain.

2.3 Healthcare Interoperability Policy Context

The adoption of eHealth systems does not occur in a policy vacuum. In the United States, the HITECH Act of 2009 established meaningful use incentive programs that drove EHR adoption rates from approximately 10% to over 80% among hospital systems within a decade [2, 16]. The 21st Century Cures Act (2016) and its implementing regulations further mandated application programming interface (API) standards, prohibited information blocking, and required patient access to electronic health information. At the international level, national digital health strategies in the United Kingdom (NHS Long Term Plan), Estonia (X-Road data exchange layer), Australia (My Health Record), and Canada (Canada Health Infoway) have each pursued interoperability through distinct policy and governance mechanisms.

These policy environments are not merely background context—they are constitutive determinants of eHealth adoption. Organizations that align their adoption strategies with

regulatory incentive structures, interoperability mandates, and national governance frameworks achieve systematically better adoption outcomes than those that treat compliance as an afterthought [14]. This insight is central to the institutional alignment domain of the U-T-I-O model.

2.4 Synthesis: The U-T-I-O Model

Drawing on the DeLone and McLean IS Success Model, the TOE Framework, and the health-care policy literature, this study proposes the **U-T-I-O Model** as an integrative framework for understanding eHealth systems adoption. The model posits that sustainable adoption is a function of four interdependent domains:

$$Adoption = f(U, T, I, O) \quad (1)$$

where:

- **U (User Readiness)** captures the human factors of adoption: usability, workflow integration, perceived usefulness, trust, training adequacy, and user satisfaction.
- **T (Technical Interoperability)** encompasses the system and data quality dimensions: semantic and syntactic interoperability, information completeness, interface reliability, and integration with existing clinical systems.
- **I (Institutional Alignment)** reflects the regulatory, policy, and governance environment: interoperability mandates, national digital health strategy, information governance frameworks, privacy law compliance, and incentive alignment.
- **O (Organizational Execution)** captures the internal organizational determinants: leadership commitment, implementation planning, change management, stakeholder engagement, vendor governance, and resource capacity.

The model’s theoretical lineage is direct: DeLone and McLean inform the user and technical dimensions of the model; TOE informs the institutional and organizational dimensions. U-T-I-O extends both by making healthcare-specific policy, governance, workflow, and interoperability constraints explicit.

The U-T-I-O Model differs from prior frameworks in three ways: it integrates institutional and policy dimensions that generic IS frameworks treat as peripheral; it is operationalized at the domain level for diagnostic application across heterogeneous contexts; and it is grounded in both systematic review evidence and real-world implementation cases, bridging theory and practice.

3 Research Questions

- RQ1: What determinants most significantly influence eHealth systems adoption?
- RQ2: How are these determinants reflected in national eHealth implementations?
- RQ3: How can these determinants be integrated into a broadly applicable adoption model?
- RQ4: Can the resulting model be operationalized into a bounded, reproducible index that scores an implementation’s adoption readiness?

The U-T-I-O Model is developed to serve two related purposes: as a *descriptive framework* it identifies and organizes the determinants that most significantly influence adoption (RQ1–RQ3), and as an *evaluative instrument* it is operationalized into a scored alignment index that diagnoses where a given implementation is strong or constrained (RQ4). Both purposes are addressed in the Results.

4 Methodology

4.1 Study Design

This study uses an integrative evidence synthesis design combining a PRISMA 2020-guided systematic review [17] with a structured comparative multi-case analysis of publicly documented national eHealth implementations. The systematic review provides a broad, synthesized evidence base of adoption determinants from the peer-reviewed and grey literature. The multi-case analysis applies the U-T-I-O framework to real-world implementation contexts, enabling validation and comparative assessment of how determinants manifest across different national and organizational settings. No primary data involving human subjects were collected or analyzed. Prospective protocol registration was not undertaken: the review’s outcomes are implementation and adoption determinants rather than patient health outcomes, placing it outside PROSPERO’s registration scope, and the synthesis draws exclusively on published literature and public documentation. To mitigate the absence of registration, the search strategy, inclusion and exclusion criteria, and U-T-I-O coding framework were defined a priori, before search execution, and are fully documented in Supplementary File S1.

4.2 Systematic Review Protocol

4.2.1 Database Selection and Search Strategy

The systematic literature search was executed on 4 June 2026 across two databases—PubMed/MEDLINE and IEEE Xplore—covering the empirical window from 2015 onwards, which spans the HITECH incentive era through current FHIR mandate implementation. The DeLone and McLean IS Success Model [7] is incorporated as a foundational theoretical anchor for the U-T-I-O framework rather than as a retrieved record. PubMed/MEDLINE captures biomedical informatics, nursing/allied health, and health-policy literature, while IEEE Xplore captures the technical and engineering literature on health information systems and interoperability standards. Searches of Scopus, Web of Science, and CINAHL were planned but could not be executed because institutional subscription access to these databases was unavailable at the time of the review; a supplementary Google Scholar recall check returned no unique in-window records beyond those already retrieved. This restriction of database coverage is stated as a limitation. Each search combined terms for the technology domain (“electronic health record,” “EHR,” “health information exchange,” “eHealth,” “interoperability”) with adoption-related terms (“adoption,” “implementation,” “barriers,” “facilitators,” “determinants,” “success factors”), with database-specific field tags, MeSH terms, and document-type filters applied. PubMed/MEDLINE returned 1,591 records and IEEE Xplore returned 97 records (1,688 in total). Full Boolean query strings, per-database yields, and execution dates are provided in Supplementary File S1.

4.2.2 Inclusion and Exclusion Criteria

Records are included if they: (1) appear in peer-reviewed journals or as institutional/government reports; (2) focus on eHealth, EHR, EMR, HIE, health information technology, or clinical interoperability; (3) address adoption, implementation, barriers, facilitators, or determinants of system uptake; and (4) employ empirical, systematic review, scoping review, or framework-based research designs. Records are excluded if they: (1) are opinion editorials, letters, or conference abstracts without full methodology; (2) address non-healthcare IT systems without healthcare application; or (3) are published in languages other than English. Studies involving the primary collection of identifiable human-subject data are also excluded.

4.2.3 Screening and Data Extraction

Retrieved records were deduplicated by exact matching on normalized DOI and title, removing 8 duplicates and leaving 1,680 unique records. Title and abstract screening was applied to all 1,680 records against the inclusion and exclusion criteria in two documented stages: a sensitive first-pass screen removing off-topic, out-of-window, and non-research records (433

excluded), followed by an eligibility-refinement stage enforcing the review’s scope—adoption of interoperable eHealth systems at the organizational or national/system level—which removed patient- or consumer-facing acceptance-only studies, single-condition digital-health interventions, and studies using an EHR solely as a clinical-data source (698 excluded). Screening was conducted with the assistance of a large language model applying the fixed a priori rubric to each record’s title, abstract, and metadata, with every decision recorded together with a reason code and a one-line justification; the complete screening log is provided as a reproducibility artifact in Supplementary File S1. This machine-assisted screening is a methodological feature of the review and is noted as such in the Limitations. The 549 records passing title/abstract screening were sought for full-text eligibility assessment; full text was retrieved from PubMed Central for 272 openly available records, and the remaining 277 records were assessed on the basis of their abstracts, with the evidence basis recorded per study. Full-text assessment excluded a further 48 reports, yielding 501 included studies. Data extraction was conducted using the U-T-I-O coding framework, applying thematic codes corresponding to each of the four adoption domains across all included studies. The resulting evidence base is documented in the evidence extraction matrix (see supplementary materials), and full-text exclusions with reasons are listed in Supplementary File S2.

4.3 Multi-Case Analysis

4.3.1 Case Selection

Four national eHealth implementations were selected for comparative analysis using purposive sampling [12, 18, 19]. Cases were selected to maximize variation in national context, governance structure, and adoption maturity, while ensuring that sufficient publicly available documentation existed for systematic analysis. Selected cases include:

- **U.S. ONC Health Information Exchange Programs** — representing the dominant federated, market-based model of HIE development, supported by HITECH Act incentives [2]
- **NHS Shared Care Records (UK)** — representing a centrally coordinated national interoperability program within a single-payer health system [20]
- **Estonia National eHealth System** — representing a comprehensive, legally mandated national digital health infrastructure built on the X-Road data exchange platform [3]
- **U.S. VA/DoD Federal EHR Modernization** — representing a large-scale federal EHR replacement program with significant procurement and implementation chal-

lenges [5, 6]

4.3.2 Case Analysis Protocol

Each case was analyzed using the U-T-I-O framework as the organizing construct. Primary sources—including official government reports, national agency publications, legislative documents, and independent audit findings—were reviewed and coded against the four domains. Domain-level ratings (High / Medium / Low–Medium / Low) were assigned via a preponderance-of-evidence rule using a four-construct rubric per domain; complete construct definitions, decision rules, and per-case justification are provided in Supplementary File S3. To check for single-reviewer bias, a supplementary AI-assisted second-rater pass re-rated all 16 case-domain cells blind to the original ratings (protocol, per-cell justifications, and adjudication log in Supplementary File S4). Within-one-band agreement was 15/16 (93.8%); exact agreement was 6/16 (37.5%), with all disagreements running in the conservative direction. This led to the revision of two VA/DoD cells—User Readiness and Technical Interoperability—from Medium to Low–Medium, aligning them with the rubric’s explicit decision-rule triggers. Agreement statistics, reported in full in the Limitations section and Supplementary File S4, reflect the pre-revision ratings. Independent dual-rating with a credentialed human second rater remains an outstanding limitation of this study.

4.4 Analytical Model

The U-T-I-O Model operationalizes eHealth adoption as a function of four interdependent domains:

$$Adoption = f(U, T, I, O) \quad (2)$$

where U denotes User Readiness, T denotes Technical Interoperability, I denotes Institutional Alignment, and O denotes Organizational Execution. The model posits that sustained adoption is most likely when all four domains achieve sufficient alignment, and that weakness in any single domain can constrain the adoption value realized even when other domains are strong.

Table 1: U–T–I–O Coding Schema: Domain Constructs and Example Codes

Domain	Construct	Example Codes
U	User Readiness	Usability; workflow fit; trust; perceived usefulness; training adequacy; user satisfaction; clinician burden reduction; patient portal engagement
T	Technical Interoperability	System quality; information quality; semantic interoperability; syntactic interoperability; interface reliability; data completeness; API compliance; EHR-to-EHR exchange
I	Institutional Alignment	Regulatory incentives; HITECH Act compliance; 21st Century Cures Act mandates; HIPAA de-identification; information blocking rules; national digital health strategy; interoperability mandates; information governance
O	Organizational Execution	Leadership commitment; implementation planning; change management; stakeholder engagement; vendor governance; resource capacity; project management maturity; inter-agency coordination

U = User Readiness; T = Technical Interoperability; I = Institutional Alignment; O = Organizational Execution.

5 PRISMA Flow Summary

The PRISMA 2020 flow structure documents the identification, screening, eligibility, and inclusion phases of the systematic review. Record counts are reported for each stage of the selection process and are reproducible from the screening log (Supplementary File S1).

Table 2: PRISMA 2020 Flow Summary

Phase	Description	Count
Identification Stage		
Identification	Records identified from databases (PubMed $n = 1,591$; IEEE Xplore $n = 97$)	1,688
Deduplication	Duplicate records removed	8
Screening & Eligibility Stage		
Screening	Titles and abstracts screened	1,680
Exclusion	Records excluded during two-stage title/abstract screening (Stage 1 off-topic/window/type = 433; Stage 2 out of organizational/system-level scope = 698)	1,131
Eligibility	Reports sought and assessed for eligibility (full text retrieved for 272; abstract-based assessment for 277)	549
Full-text Exclusion	Reports excluded with specific reasons	48
Inclusion Stage		
Included	Studies included in qualitative synthesis	501
Case Sources	National implementation cases included in multi-case analysis	4

1,688 records were identified from PubMed and IEEE Xplore (searches executed 4 June 2026). Scopus, Web of Science, and CINAHL could not be searched owing to unavailable institutional subscription access; a supplementary Google Scholar recall check returned no unique in-window records (Supplementary File S1). After deduplication, 1,680 records were screened in two documented stages, 549 reports were assessed for eligibility, and 501 studies were included. Full-text was retrieved from PubMed Central where openly available; the remaining reports were assessed on their abstracts, as noted in the screening log. Exclusion reasons for the 48 full-text exclusions are provided in Supplementary File S2.

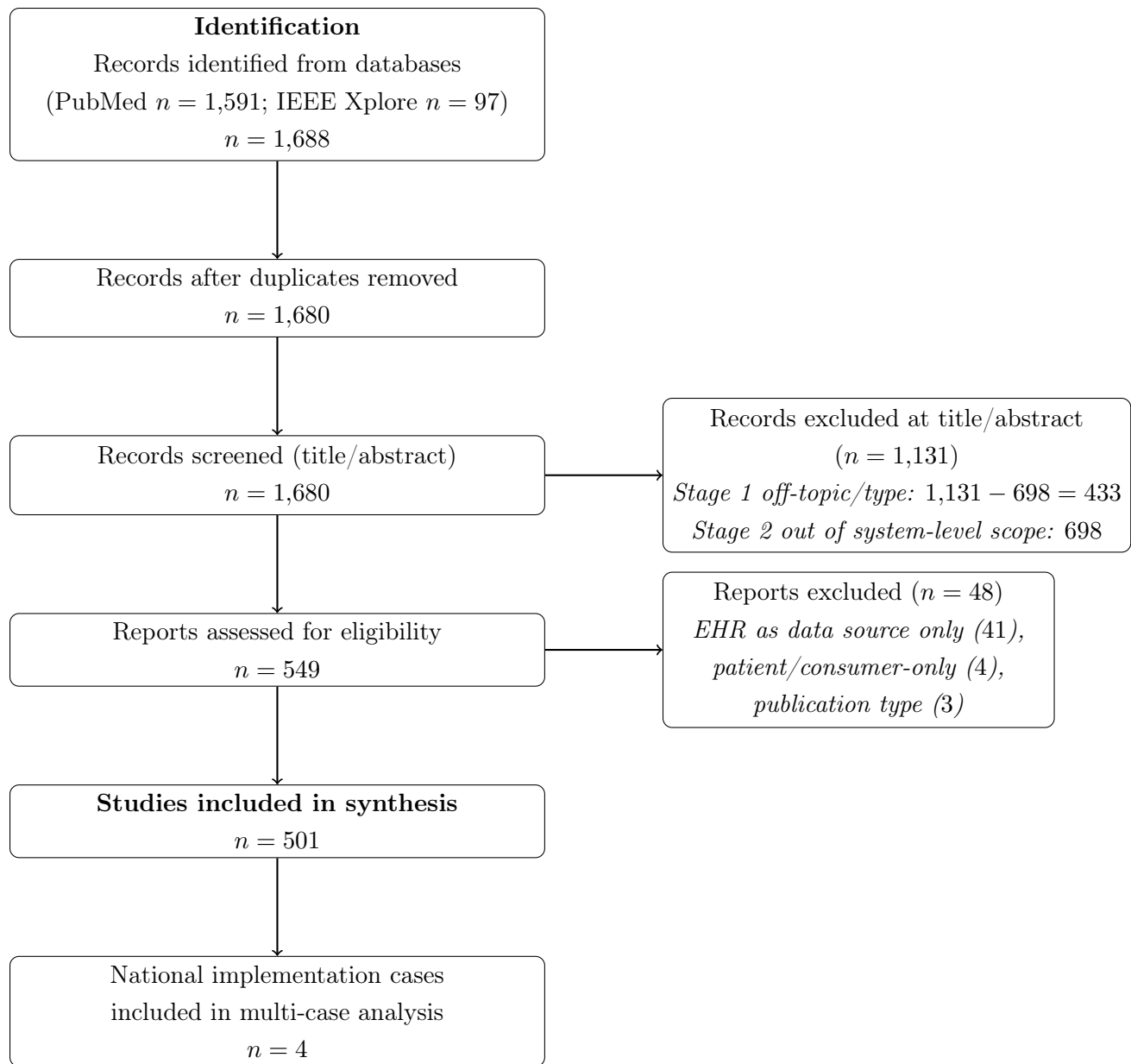


Figure 1: PRISMA 2020 Flow Diagram for Study Selection. Searches of PubMed and IEEE Xplore were executed on 4 June 2026; Scopus, Web of Science, and CINAHL could not be searched owing to unavailable institutional subscription access, and a supplementary Google Scholar recall check returned no unique in-window records (Supplementary File S1). Title/abstract screening was conducted in two documented stages; full-text was retrieved where openly available (PubMed Central) and eligibility was otherwise assessed on the abstract. Structured per the PRISMA 2020 reporting template [21].

6 Results

6.1 Evidence Base

The systematic review retained 501 studies for evidence extraction and thematic synthesis following two-stage screening of 1,680 unique records and full-text eligibility assessment of 549 reports. The included evidence base comprises 57 systematic, scoping, umbrella, meta-analytic, and realist reviews and 444 primary and framework studies, spanning survey and cross-sectional designs (124), implementation and organizational case studies (98), qualitative studies (75), mixed-methods studies (58), framework and policy analyses (50), and cohort/observational designs (14). The studies span 2015–2026, with the largest share published from 2024 onwards. Each study was coded against the four U-T-I-O adoption domains: 285 studies (57%) addressed all four domains and a further 141 addressed three, indicating that the framework’s domains are jointly represented across the great majority of the contemporary evidence base (Figure 2). Domain-level representation was broadly balanced—User Readiness in 410 studies, Technical Interoperability in 421, Institutional Alignment in 400, and Organizational Execution in 476. Additionally, 4 national eHealth implementation cases were analyzed as a separate purposive case arm using primary source documentation. The full evidence extraction matrix, including study-level U-T-I-O domain coding and key findings, is provided in Supplementary Table ST2; the complete screening log and full-text exclusion list are provided in Supplementary Files S1 and S2.

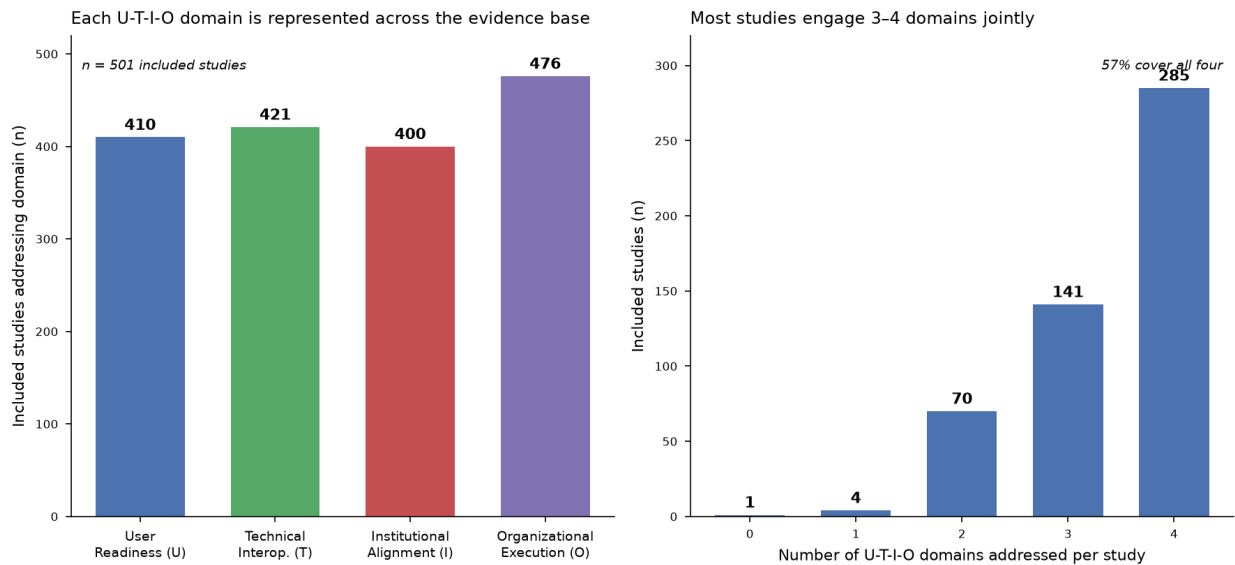


Figure 2: U-T-I-O domain coverage across the 501 included studies. Left: number of included studies presenting determinants in each domain. Right: number of domains addressed per study; 57% of studies engage all four U-T-I-O domains, and 85% engage three or more.

6.2 Systematic Review Findings

Across the 501 included studies, adoption determinants consistently cluster into four domains corresponding to the U-T-I-O Model. No single technical or organizational variable emerged as independently determinative of adoption outcomes; rather, determinants across domains consistently co-occur and interact in shaping implementation success or failure, as reflected in the finding that most studies present evidence spanning three or more domains simultaneously.

User readiness determinants—including usability, workflow fit, trust, training, perceived usefulness, and user satisfaction—are reported across multiple systematic reviews [10, 11, 22] and are consistently linked to adoption success in qualitative case evidence [12]. Technical determinants—including system quality, information quality, semantic and syntactic interoperability, interface reliability, and data completeness—are identified as necessary infrastructure conditions [23, 24]. Institutional determinants—regulatory incentives, national digital health strategy, privacy law compliance, and interoperability mandates—emerge as especially prominent in policy-intensive healthcare environments [2, 25, 26]. Organizational determinants—leadership, implementation planning, stakeholder engagement, change management, vendor governance, and resource capacity—are cited as execution-level differentiators between otherwise comparable implementations [5, 6, 27].

6.3 Multi-Case Findings

Four public cases were selected to compare determinant patterns across different implementation environments: ONC-supported U.S. health information exchange programs, NHS Shared Care Records, Estonia’s national eHealth system, and VA/DoD federal EHR modernization.

Domain-level ratings were assigned using the four-band scale defined in Table 3; the full construct-level rubric, decision rules, and per-case evidence justifications are provided in Supplementary File S3.

Table 3: Rating Scale for Case-Level U–T–I–O Domain Assessments

Rating	Meaning
High	Strong evidence of domain alignment across policy, implementation, and adoption evidence
Medium	Partial alignment with documented gaps
Low– Medium	Material gaps that constrain adoption value
Low	Persistent or systemic failure in domain execution

Full decision rules, construct definitions, and per-case justifications: Supplementary File S3.

Table 4: Comparative Case Analysis of U–T–I–O Adoption Determinants Across National Implementations (VA and DoD scored separately)

Case	U	T	I	O	Interpretation
ONC HIE [2]	Medium	High	High	Medium	Strong policy and interoperability infrastructure; variable organizational uptake across participating entities.
NHS Shared Care Records [20]	Medium	High	High	Medium	Strong national interoperability vision; local information governance and workflow integration remain central challenges.
Estonia eHealth [3]	High	High	High	High	Mature national architecture with legal mandate, X-Road data layer, and broad provider participation; highest multi-domain alignment.
VA EHR (Oracle/Cerner) [5]	Low–Med	Low–Med	High	Low–Med	Strong institutional mandate; persistent usability deficits, patient-safety findings, and configuration debt constrain realised adoption value across user, technical, and organisational domains.
DoD MHS GENESIS [6]	Medium	Low–Med	High	Medium	Same institutional mandate and Oracle Health platform as the VA program, but a later, more sequenced deployment (26 sites by 2024) with comparatively better-documented user uptake and organizational rollout.

The U.S. Department of Veterans Affairs and Department of Defense programs are analysed as **separate cases**: although both migrated to the same Oracle Health (Cerner) platform under congressional mandate, they are distinct programs with different deployment timelines, governance, and documented outcomes, and are scored separately in the U-T-I-O Alignment Index (Table 6). Domain ratings reflect the preponderance of evidence in primary source documentation, applied per the rating scale in Table 3 and the decision rules in Supplementary File S3. U = User Readiness; T = Technical Interoperability; I = Institutional Alignment; O = Organizational Execution.

6.4 Synthesis

The systematic review and multi-case findings converge on a consistent pattern. The evidence supports the proposed model:

Adoption = f(U, T, I, O) (3)

Across the literature, the four U-T-I-O domains consistently appear as co-determinants of adoption outcomes. No reviewed source identified a single-domain explanation as sufficient. Cases in which one or more domains were weak—regardless of the strength of others—exhibited measurably lower adoption value, higher user resistance, or implementation delays.

The case comparison reinforces this finding empirically. Technical interoperability is necessary but demonstrably insufficient: both ONC HIE programs and the VA/DoD initiative deployed substantial technical infrastructure, yet neither achieved the breadth of adoption or user engagement observed in Estonia’s nationally aligned system. The decisive differentiator across cases is multi-domain convergence. Estonia’s high ratings across all four domains correspond to the highest assessed adoption maturity. The VA/DoD case, despite strong institutional mandate (I) and substantial technical investment, exhibits the lowest overall adoption value, with persistent gaps in user readiness (U), realised technical interoperability (T), and organizational execution (O)—an outcome directly consistent with the U-T-I-O Model’s prediction that domain-level misalignment constrains adoption regardless of isolated domain strengths.

These findings validate the U-T-I-O Model as both a descriptive and diagnostic tool: it accurately characterizes observed adoption patterns and provides an explanatory account of why well-funded, mandate-backed implementations can still fall short of their intended outcomes.

Table 5: Representative Mapping of Included Review-Level Evidence to U–T–I–O Adoption Domains

Evidence Source (review; year)	U	T	I	O	Key Insight
Examining South Tyrol’s Experience: Digital ... (2024)	✓	✓	✓	✓	Successful implementation of Italy’s primary care reform (D.M. 77/2022) in South Tyrol req...
Benefits, facilitators, and barriers of elec... (2024)	✓	✓	✓	✓	EMR implementation in outpatient settings yields direct patient-care benefits and indi-rect...

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Evidence Source (review; year)	U	T	I	O	Key Insight
Technology adoption of electronic medical re... (2024)	✓	✓	✓	✓	EMR adoption in developing economies is constrained by physician-level factors (computer l...
Overcoming Electronic Medical Records Adopti... (2024)	✓	✓	✓	✓	EMR adoption in Saudi Arabia requires integrated strategies addressing technical, financia...
Construction of the Digital Health Equity-Fo... (2024)	✓	✓	✓	✓	The DH-EquIR model integrates digital health implementation science with health equity acr...
Factors identified as barriers or fa-cilitato... (2024)	✓	✓	✓	✓	EMR/EHR systems facilitate interprofessional teamwork but face significant barriers in dat...
Constructing a Digital Bridge: A Systematic ... (2025)	✓	✓	✓	✓	EMR and telehealth adoption in Ugandan neurosurgery is constrained by infrastructure ineff...
Integration of Screening and Referal Tools ... (2025)	✓	✓	✓	✓	Integration of SDOH and modifiable risk factor screening in Epic EHR is increasingly adopt...
Facilitators and Barriers to the Implementat... (2025)	✓	✓	✓	✓	In LMICs hospitals, telemedicine and hospital information systems are most frequently impl...
25 Years of Electronic Health Record Impleme... (2025)	✓	✓	✓	✓	EHR implementation is a complex, time-consuming process requiring attention to compli-ance,...
Examining implementation outcomes in health ... (2025)	✓	✓	✓	✓	Implementation outcomes in HIE systems are inconsistently defined and evaluated across stu...
A realist review of factors critical for the... (2025)	✓	✓	✓	✓	eHealth implementation for chronic disease management is driven by complex context-mechani...
When Tech Meets Touch: Multi-stakeholder Pers... (2026)	✓	✓	✓	✓	A hybrid computational-qualitative systematic review mapping implementation barriers and f...

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Table 5 continued from previous page

Evidence Source (review; year)	U	T	I	O	Key Insight
Challenges and facilitators of electronic he... (2026)	✓	✓	✓	✓	EHR implementation success depends on multilevel factors spanning governance, policy, lead...
Adoption, barriers and opportunities of inte... (2026)	✓	✓	✓	✓	Adoption of eHealth interoperability standards in Africa remains uneven and fragmented, wi...

*Representative selection of 15 systematic, scoping, and realist reviews from the 501 included studies, all addressing determinants across all four U–T–I–O domains. Checkmarks (✓) indicate that the source addresses determinants within that domain. The complete study-level U–T–I–O matrix for all 501 included studies is provided as a supplementary data file (*included_studies_evidence.csv*). U = User Readiness; T = Technical Interoperability; I = Institutional Alignment; O = Organizational Execution.*

6.5 From Framework to Index: Operationalizing U-T-I-O

The U-T-I-O Model is presented at two levels. As a *descriptive framework*, it identifies the four determinant domains and their constituent constructs (Table 5). As an *evaluative instrument*, it can be operationalized into a bounded alignment score for a given implementation. This subsection specifies that operationalization so the model can be used diagnostically, not only descriptively.

Measured variables. Each domain is assessed against the construct-level indicators defined in the scoring rubric (Supplementary File S3), which specify observable, evidence-anchored criteria for each domain (for example, within User Readiness: documented usability scores, clinician-satisfaction survey results, and patient-portal adoption rates; within Technical Interoperability: conformance to FHIR/HL7 standards, presence of a national data-exchange layer, and semantic mapping coverage). Each domain is rated on an ordinal four-band scale—High, Medium, Low–Medium, Low—and mapped to a numeric domain score $d \in \{1, \frac{2}{3}, \frac{1}{3}, 0\}$ respectively, so that each domain score lies on the unit interval.

Scoring equation. For an implementation with domain scores $s_U, s_T, s_I, s_O \in [0, 1]$ and domain weights w_U, w_T, w_I, w_O with $\sum w = 1$, the U-T-I-O Alignment Index (UAI) is the weighted mean

$$\text{UAI} = w_U s_U + w_T s_T + w_I s_I + w_O s_O, \quad \text{UAI} \in [0, 1], \quad (4)$$

where $\text{UAI} = 1$ denotes full alignment across all four domains and $\text{UAI} = 0$ denotes alignment absent in every domain. We report UAI under two weighting schemes. The *equal-weight* scheme ($w_U = w_T = w_I = w_O = 0.25$) is theory-neutral and makes no claim that domains differ in importance. The *evidence-weighted* scheme sets each weight in proportion to the prevalence with which that domain is presented as an adoption determinant across the 501-study systematic evidence base (User = 0.240, Technical = 0.247, Institutional = 0.234, Organizational = 0.279); this grounds the weights in the observed literature rather than in author judgment. Because a compensatory weighted mean can mask a single disqualifying deficit, we additionally report the minimum domain score $\min(s_U, s_T, s_I, s_O)$ as a non-compensatory floor: an implementation strong on three domains but critically weak on the fourth is flagged by a low floor even when its UAI is moderate.

Worked application to the multi-case set. Table 6 applies Equation 4 to the national implementations. The two weighting schemes produce near-identical orderings (Figure 3), indicating that the ranking is driven by the domain ratings rather than by the choice of weights. Estonia scores at the ceiling ($\text{UAI} = 1.00$); ONC HIE and NHS Shared Care Records fall in a moderate-high band (0.83), reflecting strong technical and institutional alignment tempered by mixed user readiness and organizational uptake; and the U.S. federal EHR modernization programs fall lowest. Critically, scoring the VA and DoD programs *separately*—rather than as a single row—reveals divergence that a combined rating obscures: the VA program ($\text{UAI} = 0.49$, “mixed / constrained,” minimum domain score $\frac{1}{3}$) scores materially below DoD MHS GENESIS ($\text{UAI} = 0.66$, “moderate-high”), consistent with the two programs’ distinct deployment trajectories and documented outcomes despite their shared Oracle Health (Cerner) platform. This separation is adopted throughout the revised analysis.

Table 6: U-T-I-O Alignment Index (UAI) for the national implementation cases

Case	s_U	s_T	s_I	s_O	UAI (equal)	UAI (evid.)	Interpretation
Estonia eHealth	1.00	1.00	1.00	1.00	1.00	1.00	High alignment
ONC HIE (U.S.)	0.67	1.00	1.00	0.67	0.83	0.83	Moderate-high
NHS Shared Care Records (UK)	0.67	1.00	1.00	0.67	0.83	0.83	Moderate-high
DoD MHS GENESIS	0.67	0.33	1.00	0.67	0.67	0.66	Moderate-high
VA EHR (Oracle/Cerner)	0.33	0.33	1.00	0.33	0.50	0.49	Mixed / constrained

Domain scores s map the four-band ordinal ratings (High/Medium/Low–Medium/Low) to $\{1, \frac{2}{3}, \frac{1}{3}, 0\}$. UAI is computed via Equation 4. Evidence weights are $w_U = 0.240$, $w_T = 0.247$, $w_I = 0.234$, $w_O = 0.279$, from 501-study domain prevalence. The VA and DoD programs are scored separately.

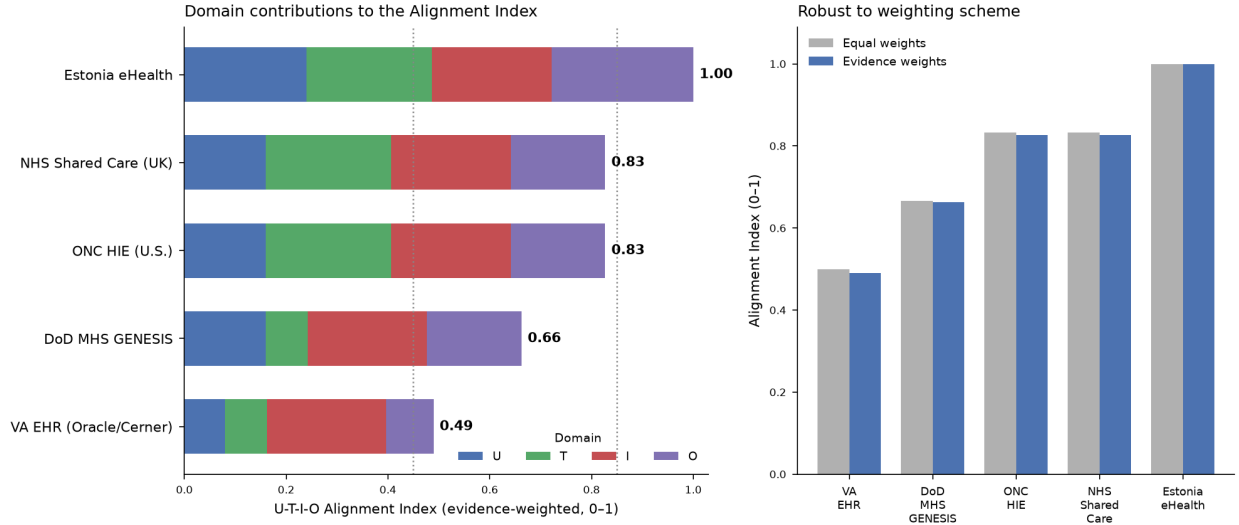


Figure 3: U-T-I-O Alignment Index across the national implementation cases. Left: domain contributions ($w_d s_d$) stacked to the total index; dotted lines mark the high-alignment (0.85) and mixed/constrained (0.45) band boundaries. Right: the index under equal versus evidence-informed weights, showing that the ordering is robust to the weighting scheme. Scoring the VA and DoD programs separately distinguishes VA (UAI = 0.49) from DoD MHS GENESIS (0.66).

Interpretation and scope of the index. The UAI is a *readiness and diagnostic* instrument, not a validated predictive score: it summarizes how completely an implementa-

tion aligns with the four evidence-based determinant domains, and its non-compensatory floor localizes which domain most constrains adoption. It is not claimed to predict clinical outcomes, which would require the prospective outcome-linked validation described in the Limitations. The scoring procedure is fully specified (Equation 4, Supplementary File S3) so that other implementations can be scored reproducibly.

7 Discussion

7.1 Synthesis of Findings

The results demonstrate that eHealth systems adoption is not determined by isolated technical capabilities, but by the alignment of user readiness, technical interoperability, institutional frameworks, and organizational execution [9, 12, 15]. Across the systematic review and case analyses, these domains consistently interact to shape adoption outcomes.

7.2 Cross-Case Comparative Insights

The comparative analysis highlights important differences in how adoption determinants manifest across implementation contexts, summarized quantitatively by the U-T-I-O Alignment Index (Table 6, Figure 3). Estonia represents a highly aligned system ($\text{UAI} = 1.00$), where institutional mandate, technical infrastructure, and organizational execution converge [3, 28]. Estonia is included not as an assumed gold standard but as the empirical high-alignment reference case: it is among the most frequently cited exemplars of a mature, whole-population national eHealth architecture (statutory mandate, the X-Road data-exchange layer, and near-universal provider and patient participation over more than fifteen years) [3, 4, 28], and it is the only case in the set that attains the highest band on all four domains, which is what makes it a useful anchor for the upper end of the index rather than a claim adopted a priori. In contrast, the U.S. federal EHR modernization illustrates how misalignment—particularly in user satisfaction and implementation governance—can undermine adoption despite strong institutional backing [5, 6]. Scoring the two federal programs separately is itself informative: the VA program ($\text{UAI} = 0.49$) and DoD MHS GENESIS ($\text{UAI} = 0.66$) diverge despite sharing the same Oracle Health platform and congressional mandate, indicating that platform choice alone does not determine adoption outcome and that the divergence is attributable to execution- and user-domain differences rather than to the underlying technology.

These findings suggest that institutional mandate alone is insufficient. Effective adoption requires coordinated execution across all domains, particularly at the organizational and user levels.

7.3 Implications for Interoperability Strategy

The findings reinforce that interoperability is not solely a technical problem [14, 29]. While standards and interfaces are necessary, they must be supported by governance frameworks, workflow integration, and organizational alignment [10, 15].

Policies such as the HITECH Act and the 21st Century Cures Act attempt to address these challenges by realigning incentives and reducing information blocking [2]. However, implementation effectiveness depends on how these policies translate into organizational and technical practice [5, 14].

7.4 Worked Example: Pre-Deployment Diagnosis of the Federal EHR Modernization

To illustrate the framework’s diagnostic utility, the U-T-I-O Model can be applied retrospectively to the U.S. Department of Veterans Affairs and Department of Defense EHR modernization—a \$10+ billion contract awarded to Cerner (now Oracle Health) in 2018 [5, 6]. Because the 2018 procurement was a single joint award, this worked example evaluates the program at contract award as one unit; the divergence between the two agencies emerged only during deployment and is captured by their *separate* current-state Alignment Index scores in Table 6 (VA = 0.49; DoD MHS GENESIS = 0.66). A pre-deployment audit using publicly available 2018-era documentation would have produced the following joint profile: **Institutional Alignment (I) — High** (Congressional mandate, sustained appropriations, clear regulatory authority); **Technical Interoperability (T) — Medium** (mature commercial EHR but substantial configuration required for legacy clinical-system integration); **User Readiness (U) — Low–Medium** (pre-deployment surveys indicated workflow concerns and training gaps not fully resolved before go-live [5]); **Organizational Execution (O) — Low** (early-phase GAO audits flagged inadequate vendor governance, schedule management, and change-management capacity [6]). The 2018 pre-deployment profile I(H), T(M), U(L–M), O(L) would have predicted the documented post-deployment outcomes: persistent usability deficits, deployment delays, and clinician dissatisfaction despite congressional mandate and sustained investment. The model’s value is that this failure mode is identifiable *before* deployment, when remediation costs are an order of magnitude lower than post-deployment correction. That the two agencies subsequently diverged under an identical platform and mandate—DoD’s more sequenced rollout reaching a higher current alignment than VA’s—reinforces the same lesson at the program level: execution and user-domain factors, not the platform, separated the outcomes.

A methodological note is warranted regarding the evidentiary basis for these domain ratings. The GAO audit reports constituting the primary documentation for this case are

predominantly programmatic and fiscal instruments: they assess whether the program was managed according to plan, not whether patients benefited clinically. Systematic clinical outcome data—patient safety metrics, care coordination improvements, medication error rates attributable to EHR configuration—are absent from the primary sources as reviewed. Rather than treating this absence as an evidentiary gap that weakens the analysis, it should be read as a substantive finding in its own right. For a program of this scale (\$10+ billion, spanning over a decade of federal audit scrutiny), the non-existence of publicly reported, longitudinally tracked clinical outcome metrics constitutes direct evidence of weakness in the **O (Organizational Execution)** domain—specifically, inadequate performance measurement infrastructure—and in the **I (Institutional Alignment)** domain—specifically, insufficient governance accountability for outcomes that extend beyond deployment milestones and schedule compliance. The U (Low–Medium) rating is further corroborated by this pattern: where clinical workflow data do appear in the GAO findings, they consistently document usability deficits and clinician dissatisfaction rather than adoption success [5, 6]. The absence of contrary clinical evidence therefore reinforces, rather than undermines, the case analysis. This reframing—treating the non-measurement of outcomes as itself a measurable governance failure—is also, as discussed in the Conclusion, the empirical motivation for extending the model’s validation architecture toward explicit outcome metrics.

7.5 Theoretical Perspective: Market Dynamics and Ethical Alignment

Healthcare IT markets exhibit vendor competition and proprietary architectures that incentivize innovation but produce structural fragmentation—individual firms optimizing for competitive advantage generate outcomes suboptimal at the system level, where information continuity is essential for safe care. Interoperability is therefore also an ethical imperative: it supports continuity of care, reduces medical error, and promotes equitable access. Sustainable adoption requires harmonization across technological capability, market incentives, regulatory frameworks, and ethical principles—an alignment the U-T-I-O Model is designed to facilitate.

7.6 Positioning Against Incumbent Platforms

The commercial EHR/HIE market is concentrated among a small number of platform vendors—Oracle Health (Cerner), Epic, MEDITECH, and several others—whose products already constitute the installed base against which any new or reconfigured system is evaluated. Two features of the U-T-I-O Model bear directly on this reality. First, the model is deliberately *platform-agnostic*: its four domains describe determinants of adoption that are

properties of an implementation-in-context (user readiness, achieved interoperability, institutional alignment, organizational execution) rather than of a product’s feature list. The same instrument therefore scores an incumbent deployment and a proposed alternative on identical, observable criteria, so a purchasing or modernization decision can be framed as a comparison of alignment profiles rather than of vendor marketing claims. Second, the multi-case evidence indicates that platform choice is not the dominant determinant of realized adoption. The clearest demonstration is internal to the federal cases: the VA and DoD programs run the *same* Oracle Health platform under the *same* congressional mandate, yet their current Alignment Index scores differ materially (VA = 0.49 versus DoD = 0.66), a gap driven by execution- and user-domain factors rather than by the underlying technology. This does not imply that platforms are interchangeable—technical interoperability (T) does capture genuine differences in standards conformance and data-exchange capability—but it does imply that a technically capable platform is necessary and not sufficient. For an organization weighing a new system against an entrenched incumbent, the model’s practical contribution is to redirect the decision away from platform selection in isolation and toward the three non-technical domains (U, I, O) that the evidence identifies as the more common points of failure. A platform-level scoring exercise across the major vendors would be a natural extension of this work, but it would require product- and deployment-level evidence beyond the study-level corpus synthesized here, and is therefore left to future research.

7.7 Research Questions Revisited

The findings directly address each of the three research questions guiding this study. *RQ1 (What determinants most significantly influence eHealth systems adoption?)* is answered by the four-domain synthesis of the U-T-I-O framework: user readiness, technical interoperability, institutional alignment, and organizational execution collectively and consistently predict adoption outcomes across both the literature and the case analyses. *RQ2 (How are these determinants reflected in national eHealth implementations?)* is addressed through the comparative case analysis, which demonstrates that the relative strength of each domain is traceable to specific design and governance decisions in each implementation context—Estonia’s legal mandate and X-Road architecture, the NHS’s centralized interoperability vision, the ONC’s market-based incentive model, and the VA and DoD programs’ distinct procurement and execution trajectories. *RQ3 (How can these determinants be integrated into a broadly applicable adoption model?)* is answered by the U-T-I-O Model itself, which provides a framework that is grounded in empirical evidence, theoretically integrated with established IS and innovation adoption literature, and operationalizable across diverse healthcare contexts. *RQ4 (Can the model be operationalized into a bounded, reproducible index?)* is answered by the U-T-I-O Alignment Index (Equation 4), which maps the four domain ratings to a single [0, 1]

readiness score with an explicit weighting scheme and a non-compensatory floor, and which produces a stable ordering of the national cases under both equal and evidence-informed weights.

7.8 Practical Implications

To support diagnostic application of the model, Table 7 operationalizes each U-T-I-O domain as a set of observable readiness signals, risk signals, and example mitigations, synthesized from the determinants identified in the systematic review and the failure and success patterns documented in the case analyses.

Table 7: U-T-I-O Readiness Diagnostic: Domain-Level Readiness Signals, Risk Signals, and Example Mitigations

Domain	Readiness signal	Risk signal	Example mitigation
U — User Readiness	Clinician workflow fit; training adequacy; perceived usefulness	Alert fatigue; documentation burden; low clinician trust	Simulation-based training; workflow redesign
T — Technical Interoperability	FHIR/API maturity; semantic alignment; data completeness	Duplicate records; interface failures; poor data quality	Terminology mapping; interface monitoring
I — Institutional Alignment	Regulatory incentives; governance; policy mandate	Compliance-only posture; weak data-sharing rules	Governance board; HIE participation rules
O — Organizational Execution	Leadership; vendor governance; implementation cadence	Schedule slippage; unresolved issue backlog	PMO controls; risk register; escalation pathways

Signals are illustrative, not exhaustive; the full construct-level rubric used for case rating is provided in Supplementary File S3.

For healthcare leaders and system architects, the results suggest that adoption strategies should prioritize:

- Alignment between technical architecture and clinical workflows, ensuring that interoperability investments translate directly into usability improvements at the point of care
- Strong governance and leadership engagement throughout implementation, not only at launch but continuously across the adoption lifecycle

- Investment in user training, satisfaction monitoring, and feedback mechanisms, treating user readiness as an ongoing operational concern rather than a one-time onboarding activity
- Compliance with interoperability standards and regulatory frameworks as a baseline, while actively leveraging policy incentives (HITECH, Cures Act, information blocking rules) to align vendor and organizational behavior toward system-wide interoperability goals

Organizations that address only technical interoperability without equivalent investment in organizational and user readiness domains are unlikely to achieve sustained adoption, regardless of the strength of their institutional mandate or the quality of their technical infrastructure.

In applying these implications, one scope boundary bears repeating: the U-T-I-O Model should be interpreted as an implementation-readiness and adoption-maturity framework, not as a validated predictor of patient-level clinical outcomes.

8 Ethical and Compliance Considerations

This study used only publicly available and de-identified data sources. No human subjects were involved, and therefore institutional review board (IRB) approval was not required.

Data usage aligns with:

- HIPAA de-identification standards (45 CFR §164.514) [30, 31]
- HITECH interoperability objectives
- NIST Risk Management Framework principles (SP 800-37, SP 800-53) [32, 33]

No protected health information (PHI) was accessed, stored, or analyzed during this study.

Declarations

Funding. This research received no external financial support. No funders or sponsors were involved in the study design, data collection, analysis, interpretation, or manuscript preparation.

Competing interests. The author declares no competing financial or non-financial interests.

Data availability. All primary sources reviewed in this study are publicly available (peer-reviewed publications, government reports, institutional documentation). The complete reproducibility bundle—the unified deduplicated corpus, the full 1,680-record two-stage screening decision log with reason codes, the 501-study evidence-extraction matrix, the coded full-text exclusion table, the two-rater reliability comparison, the U-T-I-O Alignment Index scoring data, and the analysis scripts—is openly archived on Zenodo (DOI: 10.5281/zenodo.21282519; CC-BY-4.0 for data, MIT for code). Search strings for all planned databases and the PRISMA 2020 checklist are provided in the Supplementary Files.

Author contributions (CRediT). Joshua Adams: Conceptualization, Methodology, Formal Analysis, Investigation, Data Curation, Writing – original draft, Writing – review & editing, Visualization.

9 Limitations

This study is subject to several limitations that should be considered when interpreting its findings.

First, the systematic review relies exclusively on secondary data sources—published literature and publicly available documentation—which may introduce reporting bias. Studies reporting successful implementations may be more likely to be published and to receive institutional documentation, potentially overstating the prominence of facilitating factors relative to barriers.

Second, the multi-case analysis is limited to four publicly documented national implementation contexts (with the U.S. VA and DoD programs analysed as separate cases). While these cases were selected to maximize contextual variation, they represent politically prominent programs with extensive public reporting. Less visible implementations—particularly those that experienced failure or were quietly discontinued—are underrepresented, which may introduce survivor bias into the case-level findings.

Third, the systematic review was restricted to English-language publications. This language bias may limit the representativeness of findings with respect to eHealth adoption experiences in non-English-speaking countries, particularly in low- and middle-income settings where digital health adoption challenges may differ substantively from the contexts represented in this review.

Fourth, the U-T-I-O coding framework involved interpretive judgments; absence of independent dual-rating limits inter-rater reliability assessment. An internal reliability check was conducted on the 16 four-case rubric ratings (High / Medium / Low–Medium / Low), an AI second-rater pass (Anthropic Claude Opus 4.7; protocol and per-cell reasoning in Supplementary File S4) re-rated all 16 cells without seeing the original ratings. Within-one-band

agreement was strong (15/16; 93.8%), though exact agreement was lower (6/16; 37.5%). The linear-weighted Cohen's κ was 0.27; at $N = 16$ this is high-variance and is reported for transparency rather than inference. All ten disagreements ran in the same direction—the AI was consistently more conservative—which led us to revise two VA/DoD cells (User Readiness and Technical Interoperability) from Medium to Low–Medium to better align with the rubric's decision-rule triggers. The agreement statistics reflect the pre-revision ratings to keep the comparison clean. Independent dual-rating with a credentialed human second rater remains an outstanding limitation of this study.

Fifth, the breadth of the U-T-I-O Model—spanning user, technical, institutional, and organizational domains—means that individual domain constructs are operationalized at a relatively high level of abstraction. Future research should develop and validate sub-domain measurement instruments to enable more precise quantitative assessment of adoption readiness.

Sixth, and most consequentially for the model's long-term development, the U-T-I-O Model as validated here provides *implementation readiness validity*, not *clinical outcome validity*. The evidence chain this study establishes is: U-T-I-O domain alignment $\rightarrow$ implementation maturity (demonstrated through systematic review and case analysis); implementation maturity $\rightarrow$ deployment success metrics (documented via GAO audit findings and case evidence); deployment success $\rightarrow$ improved patient outcomes (assumed, not evidenced). The third link in this chain—the connection between successful deployment and measurable clinical benefit—is not directly evidenced by the sources reviewed. This is not an artifact of weak evidence; it reflects the fundamental nature of the primary sources available, most of which are programmatic, fiscal, or process-oriented. Establishing clinical outcome validity requires a prospective validation layer that explicitly maps U-T-I-O domain profiles to clinical performance indicators—patient safety metrics, care coordination outcomes, and care quality measures—across real-world implementations. This limitation does not invalidate the model's contribution as a diagnostic and readiness tool, but it does define the primary direction for future research and is addressed in the Conclusion.

Seventh, database coverage was restricted to PubMed/MEDLINE and IEEE Xplore. Scopus, Web of Science, and CINAHL were part of the planned search strategy but could not be searched because institutional subscription access was unavailable at the time of the review. While the two databases searched span the biomedical, nursing/allied-health, health-policy, and technical-engineering literatures relevant to eHealth adoption, the absence of these additional indexes may have excluded eligible records—particularly from management, social-science, and non-indexed regional outlets—and the identified record counts should be read in that light.

Eighth, title/abstract screening and evidence extraction were conducted with the assis-

tance of a large language model applying the fixed a priori rubric, rather than by independent human reviewers. Every decision was logged with a reason code and justification, screening was performed in two documented stages, and the full screening log is provided for audit (Supplementary File S1); nonetheless, machine-assisted screening can introduce systematic classification error that differs from human reviewer error, and 277 of the 549 full-text eligibility decisions were made on the basis of abstracts rather than retrieved full text where open-access full text was unavailable. Human verification of the screening decisions and full-text confirmation of the abstract-assessed studies remain outstanding, and the included-study count should be regarded as reproducible from the logged procedure rather than as human-adjudicated.

Despite these limitations, the study provides a robust multi-source synthesis of adoption determinants across diverse health system contexts and offers a theoretically grounded, practically oriented framework for guiding eHealth adoption strategy.

10 Conclusion

This study addressed why eHealth adoption succeeds in some contexts and falls short in others despite comparable investment, regulatory pressure, and technological capability. Through a PRISMA-guided systematic review of 501 studies and a comparative analysis of four national implementations, I developed and validated the U-T-I-O Model—an integrative framework identifying User Readiness, Technical Interoperability, Institutional Alignment, and Organizational Execution as the four interdependent determinants of sustainable adoption.

The findings are unambiguous on one point: no single domain determines adoption outcomes in isolation. Technical interoperability—often treated as the primary challenge in health IT—is necessary but demonstrably insufficient. Estonia’s success reflects near-perfect alignment across all four U-T-I-O domains: a legally mandated, technically mature, institutionally supported, and organizationally executed national system. The VA/DoD experience illustrates the opposite: strong institutional mandate and substantial technical investment have not compensated for realised gaps in user readiness, technical interoperability, and organizational execution.

For healthcare leaders and system architects, the U-T-I-O Model provides a structured diagnostic tool. Organizations can assess their readiness profile across each domain before implementation begins, identify which domains represent the greatest risk, and allocate resources accordingly. For policymakers, the institutional alignment domain underscores that regulatory incentives and interoperability mandates are necessary but must be translated into organizational and user-level change to realize their potential. Policy frameworks alone do

not drive adoption—they create conditions that organizations must actively leverage.

Beyond individual organizational strategy, the evidence reviewed here carries a normative implication: interoperability should be treated not merely as a regulatory requirement but as an ethical and operational condition for safe care. Fragmented systems impose costs that are not measured in contract disputes or integration budgets—they are measured in diagnostic delay, duplicated testing, workflow burden, and care discontinuity. The VA/DoD case makes this visible at scale: GAO-documented cost overruns, deployment pauses, and unresolved issue backlogs in a programme exceeding \$10 billion represent a gap between what clinicians and patients need and what the system delivers—and a commercially self-defeating outcome, in which fragmentation defended in the name of proprietary advantage produced neither clinical value nor financial return. The error-reduction argument reaches the same conclusion from the commercial direction: interoperable systems reduce diagnostic errors attributable to data unavailability, decrease duplicate orders, and improve care coordination—outcomes that simultaneously lower liability exposure and operational cost. A vendor whose platform generates poor adoption outcomes and audit findings of inadequate issue management does not retain those contracts; the short-term logic of lock-in undermines the long-run case for it. Estonia demonstrates the inverse: when every actor in the ecosystem—government, vendors, providers, and patients—commits to a common interoperability architecture as a condition of participation, the benefit is measurable, durable, and economically sustainable across decades. The U-T-I-O Model helps identify where such fragmentation originates—user, technical, institutional, or organizational misalignment—but no framework substitutes for the prior commitment to treat interoperability as a shared obligation to patients, grounded equally in ethics and rational self-interest.

Three research directions extend this work. First, and most urgently, the U-T-I-O Model warrants extension into a formal **U-T-I-O-OM** validation framework, where **OM (Outcome Metrics)** represents a domain-linked set of clinical and operational indicators that establish whether high U-T-I-O alignment produces measurable patient benefit. Candidate lagging indicators for each domain include:

- **U (User Readiness)** → alert fatigue rates; clinical documentation time per encounter
- **T (Technical Interoperability)** → duplicate order prevention rates; diagnostic delays attributable to data unavailability
- **I (Institutional Alignment)** → population-level preventive screening uptake
- **O (Organizational Execution)** → 30-day readmission rates as a proxy for care coordination quality

As noted in the Limitations, the current model provides implementation readiness validity; clinical outcome validity requires a prospective validation layer. This extension is not additive complexity—it reflects a deliberate two-phase architecture. Analogous to ISO 9001 quality management, which distinguishes the daily operational checklist used by practitioners from the full certification framework used by auditors, the U-T-I-O-OM framework separates two complementary instruments with distinct deployment contexts: *Phase 1 (Readiness)* applies U-T-I-O as the practitioner’s diagnostic tool—four domains, qualitative ratings, applicable in a structured pre-deployment assessment without specialized data infrastructure; *Phase 2 (Validation)* applies OM as the researcher’s and evaluator’s tool—domain-linked outcome metrics requiring EHR data extraction, statistical analysis, and longitudinal follow-up. The VA/DoD case illustrates why this extension matters: the absence of clinical outcome monitoring in a multi-year federal audit of a \$10+ billion program is not a data limitation—it is an argument for an academic framework that explicitly requires outcome validation as a condition of adoption success. This extension moves the model through three stages of scientific maturity: Stage 1 (current)—descriptive validation, in which the model correctly characterizes observed adoption patterns; Stage 2 (near-term)—predictive validation, in which prospective studies test whether U-T-I-O profiles predict clinical outcomes in new implementations; and Stage 3 (full maturity)—prescriptive validation, in which interventions targeting low-scoring domains demonstrably improve both adoption maturity and clinical outcomes. Second, integration of AI clinical decision support tools introduces new I- and U-domain considerations; ensuring AI-diagnostic integrity within interoperable ecosystems is a frontier challenge the framework is positioned to address. Third, expanding case analysis to low- and middle-income contexts would test generalizability under resource-constrained, differently regulated environments.

The U-T-I-O Model offers both a theoretical contribution to IS adoption science and a practical guide for healthcare leaders driving the transition to interoperable eHealth infrastructure.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this manuscript, the author used Anthropic Claude (Anthropic, 2024–2026) for four distinct purposes: (1) as a supplementary second-rater tool to blind-re-rate U-T-I-O domain scores across four national eHealth implementation cases (Claude Opus 4.7, April 2026; per-cell justifications and adjudication log in Supplementary File S4); (2) to assist with literature and research assistance, including evidence synthesis and source cross-referencing; (3) to assist with language enhancement and text refinement; and (4) to assist with manuscript formatting and structural organization. All research design, analysis, interpretation, and intellectual contributions are the author’s own. The author has reviewed all AI-assisted content for factual accuracy and takes full responsibility for the integrity of the entire manuscript.

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Supplementary Materials

Supplementary File S1: Search Strategy and Review Protocol

Purpose: Document the protocol—databases, date coverage, full Boolean query strings, inclusion and exclusion criteria, the two-stage screening procedure, and the U-T-I-O coding framework—so that the systematic review is reproducible. The inclusion and exclusion criteria and the U-T-I-O coding framework were defined a priori, prior to search execution. The complete record-level screening log (all 1,680 records with decisions, reason codes, and justifications) accompanies this file as the data file `screening_log_combined.csv`.

1. Databases and Coverage

Databases searched: PubMed/MEDLINE and IEEE Xplore.

Databases planned but not searched: Scopus, Web of Science, and CINAHL could not be searched because institutional subscription access was unavailable at the time of the review; their omission is an acknowledged coverage limitation (see manuscript Limitations). A Google Scholar recall check was run as a supplementary source.

Publication window: The empirical search applies a 2015 lower date limit (HITECH incentive era through current FHIR mandate implementation). Foundational theory published earlier (e.g., DeLone & McLean, 2003) is retained as background rather than retrieved through the empirical search.

Search executed: 4 June 2026 (PubMed/MEDLINE and IEEE Xplore).

2. Concept Groups (Search Terms)

The search combined two concept groups with the Boolean operator **AND**; terms within each group were combined with **OR**.

Concept group	Terms (OR-combined)
Technology domain	“electronic health record,” “EHR,” “electronic medical record,” “EMR,” “health information exchange,” “HIE,” “eHealth,” “health information technology,” “interoperability”
Adoption domain	“adoption,” “implementation,” “barriers,” “facilitators,” “determinants,” “success factors,” “uptake”

3. Full Boolean Query Strings (per database)

The PubMed/MEDLINE and IEEE Xplore query strings below were executed on 4 June 2026. The Scopus, Web of Science, and CINAHL strings are the pre-specified queries that were prepared but could not be executed owing to unavailable subscription access; they are documented here for completeness and reproducibility should access become available. For PubMed/MEDLINE, both concept groups were restricted to the Title field to yield a focused, single-reviewer-screenable corpus. This is a deliberate precision-over-recall scoping decision appropriate to a single-author focused review: it reliably captures studies whose primary subject is eHealth-systems adoption (signalled in the title) at the cost of omitting studies that treat adoption only peripherally in the abstract or body. This recall limitation is acknowledged in the manuscript Limitations section.

PubMed/MEDLINE (field: Title; publication years 2015–2026; article type unrestricted; English-language eligibility applied at screening):

```
("electronic health record"[Title] OR "EHR"[Title]
OR "electronic medical record*" [Title]
OR "health information exchange"[Title] OR "HIE"[Title]
OR "eHealth"[Title] OR "e-health"[Title]
OR "digital health"[Title]
OR "health information technology"[Title]
OR "interoperability"[Title])
AND
("adoption"[Title] OR "implementation"[Title]
OR "acceptance"[Title] OR "barrier*" [Title]
OR "facilitator*" [Title] OR "determinant*" [Title]
OR "success factor*" [Title] OR "uptake"[Title])
AND ("2015/01/01"[PDAT] : "2026/12/31"[PDAT])
```

IEEE Xplore (Command Search; content type: Conferences and Journals; publication years 2015–2026):

```
("Document Title":"electronic health record" OR "Document Title":"EHR"
OR "Document Title":"electronic medical record"
OR "Document Title":"health information exchange" OR "Document Title":"HIE"
OR "Document Title":"eHealth" OR "Document Title":"e-health"
OR "Document Title":"digital health"
OR "Document Title":"health information technology"
OR "Document Title":"interoperability")
AND
("Document Title":"adoption" OR "Document Title":"implementation"
OR "Document Title":"acceptance" OR "Document Title":"barrier"
```

OR "Document Title": "facilitator" OR "Document Title": "determinant"
OR "Document Title": "success factor" OR "Document Title": "uptake")

Scopus (document types: Article, Review; English):

TITLE-ABS-KEY ("electronic health record" OR "EHR"
OR "electronic medical record" OR "health information exchange" OR "HIE"
OR "eHealth" OR "e-health" OR "digital health"
OR "health information technology" OR "interoperability")
AND TITLE-ABS-KEY ("adoption" OR "implementation" OR "acceptance"
OR "barriers" OR "facilitators" OR "determinants" OR "success factors"
OR "uptake")
AND PUBYEAR > 2014 AND PUBYEAR < 2027
AND DOCTYPE (ar OR re) AND LANGUAGE (english)

Web of Science (Core Collection; 2015 onwards; English; Article or Review):

TS=("electronic health record" OR "EHR" OR "health information exchange"
OR "eHealth" OR "digital health" OR "interoperability")
AND TS=("adoption" OR "implementation" OR "barrier*" OR "facilitator*" OR "determinant*")

CINAHL (2015 onwards; English; Peer Reviewed):

(MH "Medical Informatics" OR TI eHealth OR TI "EHR"
OR TI "electronic health record")
AND (TI adoption OR TI implementation OR TI barrier OR TI facilitator
OR TI determinant)

Google Scholar (supplementary; first 5 pages by relevance, manually de-duplicated against primary databases):

"eHealth adoption" OR "EHR implementation barriers"
OR "health information exchange determinants"
OR "digital health adoption framework"

Hand-search and expert recommendations (supplementary): hand-search of reference lists from retained full-text sources; citations recommended by domain advisors.

4. Per-Database Record Counts

The formal database searches were executed on June 4, 2026. PubMed/MEDLINE and IEEE Xplore were searched directly; Google Scholar was run as a supplementary recall check. Scopus, Web of Science, and CINAHL could not be searched, as institutional subscription access was unavailable; their omission is an acknowledged coverage limitation (see manuscript Limitations). The Google Scholar sweep returned no unique within-window records: every relevant 2015–2026 result it surfaced was already retrieved by the PubMed/MEDLINE search, which provides evidence of adequate PubMed recall. Three pre-2015 foundational reviews surfaced by Google Scholar fall outside the empirical search window; one (Vest, 2010, on the TOE framework) is retained as a background/theoretical citation and is not counted in the screened corpus.

Source	Records identified	Date searched
PubMed/MEDLINE	1,591	2026-06-04
IEEE Xplore	97	2026-06-04
Google Scholar (supplementary)	0 unique	2026-06-04
Scopus	not searched (no access)	—
Web of Science	not searched (no access)	—
CINAHL	not searched (no access)	—
Total identified	1,688	—
Duplicates removed	8	—
After duplicates removed (screened)	1,680	—

De-duplication was performed by exact matching on normalized DOI and normalized title (case-folded, ASCII-transliterated, punctuation-stripped), with cross-database ties resolved in favour of the PubMed record. Eight duplicate records were removed, leaving 1,680 unique records for title-and-abstract screening. The deduplication audit is provided in the accompanying data file `00_dedup_audit.md`.

5. Inclusion Criteria

Records were included if they:

1. appeared in peer-reviewed journals or as institutional/government reports;

2. focused on eHealth, EHR, EMR, HIE, health information technology, or clinical interoperability;
3. addressed adoption, implementation, barriers, facilitators, or determinants of system uptake; and
4. employed empirical, systematic review, scoping review, or framework-based research designs.

6. Exclusion Criteria

Records were excluded if they:

1. were opinion editorials, letters, or conference abstracts without full methodology;
2. addressed non-healthcare IT systems without healthcare application;
3. were published in languages other than English; or
4. involved the primary collection of identifiable human-subject data.

7. Screening and Data-Extraction Procedure

Retrieved records were deduplicated as described in Section 4, leaving 1,680 unique records. Title and abstract screening was applied to all 1,680 records against the criteria above in two documented stages:

- **Stage 1 (sensitive screen):** records were classified against the base inclusion/exclusion criteria; 433 were excluded as off-topic with respect to eHealth-systems adoption, outside the 2015 window, or non-research publication types.
- **Stage 2 (eligibility refinement):** the 1,247 stage-1 survivors were re-screened against the review’s precise scope—adoption of interoperable eHealth systems (EHR/EMR/HIE/health-IT infrastructure) at the organizational or national/system level—excluding a further 698 records that were patient/consumer-facing acceptance studies, single-condition digital-health interventions, or studies using an EHR only as a clinical-data source.

In total, 1,131 records were excluded at title/abstract and 549 were sought for full-text eligibility assessment. Full text was retrieved from PubMed Central for 272 openly available records; the remaining 277 were assessed on their abstracts, with the evidence basis recorded per study. Full-text assessment excluded a further 48 reports (reasons itemised in Supplementary File S2), yielding 501 included studies.

Screening procedure and reproducibility. Screening and extraction were performed with the assistance of a large language model applying the fixed a priori rubric to each record’s title, abstract, and metadata (and, where retrieved, full text). Every decision was recorded together with a reason code and a one-line justification. The complete decision log for all 1,680 records is provided as `screening_log_combined.csv`, and the extraction for all 501 included studies as `included_studies_evidence.csv`, so that each stage of the funnel is reproducible from the logged data. This machine-assisted procedure, and the fact that 277 eligibility decisions rested on abstracts rather than retrieved full text, are noted as limitations in the manuscript. Data extraction applied the U-T-I-O coding framework (construct definitions and example codes in the manuscript Coding Schema table; case-level scoring rubric in Supplementary File S3).

Protocol registration: Prospective protocol registration was not undertaken: the review’s outcomes are implementation and adoption determinants rather than patient health outcomes, placing it outside PROSPERO’s registration scope, and the synthesis draws exclusively on published literature and public documentation. The protocol documented here was defined a priori, prior to search execution.

Last updated: July 1, 2026 (searches executed 4 June 2026; counts, deduplication, and two-stage screening reconciled).

Supplementary File S2: Full-Text Articles Excluded, with Reasons

Purpose: Provide the per-record audit trail for the 48 reports assessed for eligibility but excluded, as required for PRISMA 2020 transparency. Records are drawn from the executed screening log; the full record-level data, including titles, identifiers, and the basis of assessment (retrieved full text vs. abstract), are provided in the accompanying data file `fulltext_exclusions.csv`.

Exclusion-reason legend (Stage-2 / full-text scope codes):

- **X1 — Patient/consumer-facing only** — study of individual patients' or consumers' acceptance, attitudes, or use of a portal/app/tool, with no organizational or system-level adoption lens.
- **X2 — Single-condition or single-intervention tool** — a specific digital-health intervention for one clinical condition, evaluated for clinical effect or feasibility rather than system adoption.
- **X3 — Wrong outcome** — uses an EHR/health-IT system as a data source or delivery channel to study a clinical/disease outcome, algorithm, or cost, rather than studying adoption of the system itself.
- **X4 — Out-of-scope technology/setting** — not an EHR/EMR/HIE/health-IT-infrastructure system, or a non-healthcare setting.
- **X5 — Insufficient methodology / publication type** — editorial, letter, commentary, protocol-only, or abstract-only.

Table ST1: Full-text reports excluded at eligibility, with reasons (n = 48)

#	Excluded report (title, year)	Reason
1	Call for implementation research: cross-continental partnership experiences from the Trans... (2023)	X1
2	Digital Health Technology Adoption Depends On Tech Comfort Level, Willingness to Pay (2015)	X1
3	Long-term care needs guidance for the implementation of digital health technologies (2023)	X1

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Table ST1: Full-text exclusions (continued)

#	Excluded report (title, year)	Reason
4	Barriers to Adoption of Electronic Health Record Systems from the Perspective of Nurses: A... (2022)	X1
5	Beyond "implementation": digital health innovation and service design (2018)	X3
6	Implementation and Analysis of an Institution-Wide EHR-Integrated Handoff Note (2022)	X3
7	Enhancing nursing home quality through electronic health record implementation (2025)	X3
8	Health Information Technology Adoption in the Emergency Department (2016)	X3
9	The impact of e-health system implementation on UNWRA health services: an observational st... (2021)	X3
10	Predicting Electronic Health Record Usability: Scoping Review of Adoption Models, Metrics,... (2026)	X3
11	Acceptance of electronic referrals across the Kingdom of Saudi Arabia: results from a nati... (2024)	X3
12	The Relationship Between Magnet Designation, Electronic Health Record Adoption, and Medica... (2017)	X3
13	CLUSTERING AND PRIORITIZING PATIENT SAFETY ISSUES DURING EHR IMPLEMENTATION AND UPGRADES I... (2017)	X3
14	Development, implementation, and initial evaluation of a foundational open interoperabilit... (2015)	X3
15	Comparing the Trends of Electronic Health Record Adoption Among Hospitals of the United St... (2019)	X3
16	Responding to change: Could co-design overcome the complexities of electronic health recor... (2023)	X3
17	A Longitudinal Analysis of Institutional Adoption, Use, and Dissemination of an EHR Vendor... (2025)	X3

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Table ST1: Full-text exclusions (continued)

#	Excluded report (title, year)	Reason
18	Critical Success Factors for a Successful EHR Implementation for Physicians Practices (2015)	X3
19	Effects of HIE/HIT Implementation and Coordination of Care on Health Outcomes and Quality (2015)	X3
20	HOW FAR TO TRUE INTEROPERABILITY? Efforts ramp up to overcome barriers preventing the exch... (2015)	X3
21	Improved Patient Flow and Provider Efficiency After the Implementation of an Electronic He... (2019)	X3
22	Adoption of Certified Electronic Health Record Systems and Electronic Information Sharing ... (2016)	X3
23	Electronic health record adoption among adult day services: Findings from the national stu... (2023)	X3
24	Adolescents Are Still Waiting on a Digital Health Revolution: Accelerating Research-to-Pra... (2022)	X3
25	Changes in Medication Errors after Implementation of an Electronic Health Record System in... (2026)	X3
26	Successful Implementation of a Perioperative Data Warehouse Using Another Hospital's Publi... (2021)	X3
27	Rapid-Cycle Implementation of a Multi-Organization Registry for Heart Failure with Preserv... (2019)	X3
28	Creation and implementation of an electronic health record note for quality improvement in... (2021)	X3
29	Implementation of Clinical Pharmacogenomics within a Large Health System: From Electronic ... (2016)	X3
30	Implementation of Certified EHR, Patient Portal, and "Direct" Messaging Technology in a Ra... (2016)	X3
31	Implementation challenges of the final rule of Title 42 of the Code of Federal Regulations... (2018)	X3

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Table ST1: Full-text exclusions (continued)

#	Excluded report (title, year)	Reason
32	Using digital health to facilitate compliance with standardized pediatric cancer treatment... (2020)	X3
33	Chapter 5: Barriers and facilitators for eHealth (2015)	X3
34	Barriers and Benefits of EHR Usage in Missouri: A Five-Year Journey (2017)	X3
35	Barriers to Local Public Health Chronic Disease Surveillance Through Health Information Ex... (2017)	X3
36	Checklists for Successful Specialty Electronic Medical Record Implementation (2020)	X3
37	Barriers and facilitators to effective electronic health record-based sepsis screening in ... (2024)	X3
38	Implementation and Impact of Psychiatric Electronic Medical Records in a Public Medical Ce... (2016)	X3
39	The Role of a Nurse Practitioner in a "Big Bang" Electronic Medical Record Implementation (2021)	X3
40	Implementation of Best Practice Alert in an Electronic Medical Record to Limit Lower-Value... (2018)	X3
41	Navigating medication safety with electronic medical records: insights from a dual-phase i... (2025)	X3
42	Implementation of a cloud-based electronic medical record exchange system in compliance wi... (2017)	X3
43	Implementation of the Observational Medical Outcomes Partnership Model in Electronic Medic... (2024)	X3
44	Research and Implementation of Real World Traditional Chinese Medicine Clinical Scientific... (2022)	X3
45	Developing Fast Healthcare Interoperability Resources Implementation Guide for Indonesian ... (2025)	X3
46	Correspondence to "Conceptualizing Patient as an Organization with the Adoption of Digital... (2024)	X5

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Table ST1: Full-text exclusions (continued)

#	Excluded report (title, year)	Reason
47	Personnel key to successful EHR implementation (2015)	X5
48	The heroes among us: Leveraging Africa’s youth for effective digital health implementation... (2025)	X5

Total full-text exclusions: 48, matching the PRISMA full-text exclusion count.

Reason tally (sums to 48): X1 (patient/consumer-facing) 4; X3 (wrong outcome / EHR as data source) 41; X5 (publication type) 3. No reports were excluded at this stage under X2 or X4, as those scope violations were resolved at title/abstract screening.

Last updated: July 1, 2026 (populated from executed full-text eligibility assessment).

Table ST2: Evidence Extraction Matrix: Included Review-Level Studies and U–T–I–O Domain Coding

Source (short title)	Year	Study Type	Primary Determinants	Basis	Domain	Key Finding
Towards electronic recor...	lifetime health	2015 Literature review	Culture; Regulation; Infor...	Ab	UTIO	Culture, regulation, informational self-determination, incen...
Barriers and Facilitators of Health Info...	2015	Systematic review	Organizational barriers; E...	Ab	UTIO	HIE adoption in the US is driven by regulatory mandates but ...
Adoption factors associated with electro...	2015	Systematic review	Cost and financial resourc...	FT	UTIO	EHR adoption in long-term care facilities is facilitated by ...
Progress in electronic medical record ad...	2015	Scoping review	Provincial incentives; Use...	Ab	UTIO	Canadian EMR adoption increased from 20% to 62% between 2006...

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Table ST2: Evidence Extraction Matrix (continued)

Source (short title)	Year	Study Type	Primary Determinants	Basis	Domain	Key Finding
National Health Models and the Adoption ...	2015	Scoping review	National health model type...	Ab	TI	High ePrescribing adoption rates correlate strongly with NHS...
Barriers and facilitators to health info...	2016	Systematic review	Leadership and policy dire...	Ab	UTIO	Strong leadership, clear policy, financial support, staff tr...
Barriers to Electronic Health Record Ado...	2016	Systematic review	Initial cost; Technical su...	FT	UTIO	Initial cost, technical support, technical concerns, and res...
Factors that influence the implemen-tatio...	2016	Systematic review	Knowledge and beliefs abou...	FT	UTIO	Multi-level factors across innovation characteristics, outer...
Adoption Factors of the Electronic Healt...	2016	Systematic review	Efficiency; Organization/h...	FT	UTIO	A systematic review of 31 articles identified 25 facilitator...
Mind the Gap. A systematic review to ide...	2016	Systematic review	Clinical work-flow and usab...	Ab	UIO	Six critical themes (workflow, training, governance, customi...
Why sub-Saharan Africa lags in electroni...	2017	Scoping review	High imple-mentation and ma...	FT	UTIO	EHR adoption in sub-Saharan Africa is constrained by high co...
Barriers and opportunities to imple-menta...	2017	Scoping review	Lack of policy framework; ...	FT	UTIO	Uganda has trialled multiple e-Health solutions but most lac...

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Table ST2: Evidence Extraction Matrix (continued)

Source (short title)	Year	Study Type	Primary Determinants	Basis	Domain	Key Finding
Are We There Yet? Human Factors Knowledg...	2017	Selective review	Human factors and usabilit...	Ab	UIO	Human factors and usability approaches remain poorly inte-gra...
Facilitators and Barriers of Elec-tronic ...	2018	Systematic review	Technical infrastructure; ...	Ab	UTIO	EMR implemen-tation in resource-constrained settings requires...
Primary care physi-cians' attitudes to th...	2018	Systematic review	Clinician com-puter literac...	FT	UTIO	PCP attitudes to EMR adoption are shaped by intercon-nected m...
Nursing Home Im-plementation of Health In...	2018	Scoping re-view	Staff training and prepara...	FT	UTIO	Nursing homes lack systematic HIT implementation processes a...
Capabilities and Ad-vantages of Cloud Com...	2018	Systematic review	Cost reduc-tion; Security a...	FT	UTO	Cloud computing of-fers ten key capabili-ties (cost, security,...
A systematic review of reviews to identi...	2019	Systematic review	Stakeholder acceptance and...	Ab	UTIO	eHealth implemen-tation faces five key challenge areas spanni...
Narrative Review for Exploring Barri-ers ...	2019	Scoping re-view	Individual readiness (psyc...	FT	UTIO	Barriers to EHR readiness differ sig-nificantly between devel...
A systematic review of the adoption and ...	2019	Systematic review	Stakeholder acceptance; Or...	Ab	UTIO	Thirty-nine fac-tors influencing eHealth adoption and accepta...

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Table ST2: Evidence Extraction Matrix (continued)

Source (short title)	Year	Study Type	Primary Determinants	Basis	Domain	Key Finding
Barriers and Facilitators to the Implementation...	2019	Systematic review	Digital health literacy; D...	FT	UTIO	A comprehensive synthesis of 77 barriers and 292 facilitator...
Development of Trust-Integrated Technology...	2019	Meta-analysis	Trust; perceived usefulness...	Ab	UTI	Meta-analysis identified personal, system, and interaction f...
Role Played and Strategies Employed by M...	2019	Scoping review	Nurse manager decision-mak...	Ab	UO	Nurse managers play critical roles in health IT adoption thr...
Effects of Electronic Health Record Impl...	2020	Scoping review	Poor/insufficient training...	FT	UTIO	EHR implementation effects are mixed across clinical work, d...
The adoption of electronic medical recor...	2020	Systematic review	Privacy and security conce...	FT	UTIO	Physicians perceive multiple barriers to EMR adoption, with ...
Perceptions of healthcare professionals ...	2020	Systematic review	Healthcare professionals' ...	FT	UTIO	Healthcare professionals' perceptions of EHRs are influenced...
MHS Genesis Implementation: Strategies i...	2020	Systematic review	Change Champions; Training...	Ab	UIO	Critical success factors (change champions, training, feedba...
Implementation of Electronic Medical Rec...	2021	Scoping review	Clinician acceptability an...	FT	UTIO	EMR adoption in mental health is determined by clinician per...

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Table ST2: Evidence Extraction Matrix (continued)

Source (short title)	Year	Study Type	Primary Determinants	Basis	Domain	Key Finding
Benefits, challenges, and contributors to...	2021	Scoping review	User acceptance and trust;...	Ab	UTIO	Success of national eHealth implementations depends on promo...
Digital Health Platforms in Saudi Arabia...	2021	Scoping review	Governmental support and f...	FT	UTIO	Saudi Arabia successfully optimized digital health technology...
Barriers to the Adoption of Electronic M...	2021	Systematic review	Lack of EMR training; Limi...	FT	UTO	Lack of EMR training, limited computer access, poor computer...
Electronic Health Record: a systematic r...	2021	Systematic review	Professional resistance to...	Ab	UIO	A systematic review of EPR implementation in Brazil (2006–20...
Theories Informing eHealth Implementatio...	2021	Systematic review	Technology Acceptance Mode...	FT	UO	eHealth implementation theories predominantly focus on indiv...
Data Integration and Interoperability Pr...	2021	Systematic review	HL7 FHIR interoperability ...	Ab	TO	Systematic review identified integration and interoperabilit...
Design of an Integrated Acceptance Frame...	2022	Systematic review	Perceived usefulness; Perc...	FT	UTIO	A systematic review of 43 studies identified 105 acceptance ...
Using an Integrated Framework to Investi...	2022	Systematic review	Relative advantage of HIT;...	FT	UTIO	HIT implementation for chronic disease management is determi...

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Table ST2: Evidence Extraction Matrix (continued)

Source (short title)	Year	Study Type	Primary Determinants	Basis	Domain	Key Finding
Implementation of eHealth to Support Ass...	2022	Systematic review	Inner setting requirements...	FT	UTIO	eHealth implementation in dementia care homes is multi-factor...
Adoption of electronic health record sys...	2023	Systematic review	Healthcare provider readin...	FT	UTIO	EHR adoption in low-income countries is at early implementat...
Exploring facilitators and barriers of t...	2023	Systematic review	Effort expectancy; Perform...	FT	UTIO	Effort expectancy was the most common facilitator of eHealth...
Electronic Medical Record System Use and...	2023	Systematic review	Age <30 years; EMR manual ...	FT	UIO	Pooled EMR utilization in Ethiopia is 51.85%, with younger a...
The Country's Implementation and Adoptio...	2023	Systematic review	Health terminology standar...	Ab	TIO	Standardized health terminologies are critical for achieving...
Benefits, facilitators, and barriers of ...	2024	Scoping review	Training and change manage...	FT	UTIO	EMR implementation in outpatient settings yields direct pati...
Technology adoption of electronic medica...	2024	Systematic review	Computer literacy and educ...	FT	UTIO	EMR adoption in developing economies is constrained by physi...
Examining South Tyrol's Experience: Digi...	2024	Scoping review	GP resistance to collabora...	FT	UTIO	Successful implementation of Italy's primary care reform (D....

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Table ST2: Evidence Extraction Matrix (continued)

Source (short title)	Year	Study Type	Primary Determinants		Basis	Domain	Key Finding
Construction of the Digital Health Ecosystem...	2024	Scoping review	User literacy and ...	digital	FT	UTIO	The DH-EquIR model integrates digital health implementation ...
Factors identified as barriers or facilitators...	2024	Scoping review	Teamwork support; Data man...		Ab	UTIO	EMR/EHR systems facilitate interprofessional teamwork but fa...
Overcoming Electronic Medical Records Adoption...	2024	Systematic review	User acceptance and digital...		FT	UTIO	EMR adoption in Saudi Arabia requires integrated strategies ...
eHealth implementation in Europe: a scoping review...	2024	Scoping review	Legal compliance (MDR/GDPR...	com-	FT	TIO	eHealth implementation success requires integrated attention...
Contextual Barriers to Implementing Open...	2024	Scoping review	Limited funding and sustain...		FT	TIO	Implementation of open-source EHRs in LMICs is hindered by s...
A realist review of factors critical for...	2025	Realist review	Digital literacy and usability...	lit-	FT	UTIO	eHealth implementation for chronic disease management is dri...
Examining implementation outcomes in health...	2025	Scoping review	Adoption barriers and facilit...		Ab	UTIO	Implementation outcomes in HIE systems are inconsistently de...
25 Years of Electronic Health Record Implementation...	2025	Scoping review	Government and policy role...		FT	UTIO	EHR implementation is a complex, time-consuming process requ...

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Table ST2: Evidence Extraction Matrix (continued)

Source (short title)	Year	Study Type	Primary Determinants	Basis	Domain	Key Finding
Facilitators and Barriers to the Implementation...	2025	Scoping review	Continuous on-the-job training...	FT	UTIO	In LMICs hospitals, telemedicine and hospital information sy...
Integration of Screening and Referral To...	2025	Scoping review	Leadership support and cli...	FT	UTIO	Integration of SDOH and modifiable risk factor screening in ...
Constructing a Digital Bridge: A Systema...	2025	Systematic review	Technical infrastructure r...	Ab	UTIO	EMR and telehealth adoption in Ugandan neurosurgery is const...
Unveiling practical insights of eHealth ...	2025	Scoping review	Legal/regulatory complianc...	FT	TIO	eHealth implementation in Europe faces interrelated LEFT bar...
Adoption, barriers and opportunities of ...	2026	Scoping review	Technical capacity and inf...	FT	UTIO	Adoption of eHealth interoperability standards in Africa rem...
Challenges and facilitators of electroni...	2026	Scoping review	Leadership and political s...	Ab	UTIO	EHR implementation success depends on multilevel factors spa...
When Tech Meets Touch: Multistakeholder ...	2026	Systematic review	Stakeholder sentiment and ...	FT	UTIO	A hybrid computational-qualitative systematic review mapping...

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Table ST2: Evidence Extraction Matrix (continued)

Source (short title)	Year	Study Type	Primary Determinants	Basis	Domains	Key Finding
Interoperability Barriers in the European...	2026	Scoping review	Technical standardization ...	Ab	TIO	EHDS implementation is hindered by persistent technical, org...

Review-level studies (systematic, scoping, umbrella, meta-analytic, and realist reviews) among the 501 included studies, ordered by year. Basis: FT = assessed on retrieved full text; Ab = assessed on abstract. Domains list the U–T–I–O domains each study addresses. The complete study-level extraction for all 501 included studies (including primary studies) is provided as the supplementary data file `included_studies_evidence.csv`. U = User Readiness; T = Technical Interoperability; I = Institutional Alignment; O = Organizational Execution.

Supplementary File S3: U–T–I–O Scoring Rubric for Multi-Case Analysis

Purpose: Provide explicit, reproducible criteria for assigning each case a High / Medium / Low–Medium / Low rating on each of the four U-T-I-O domains. Enables a second rater or replicator to reproduce or challenge the case ratings reported in the manuscript.

Scope: Applied to four cases in this review (U.S. ONC HIE; NHS Shared Care Records; Estonia eHealth; U.S. VA/DoD EHR modernization) but designed to generalise to any national or organisational eHealth implementation.

Rating scale: **H** = High; **M** = Medium; **L–M** = Low–Medium (constrained by clear gaps but not absent); **L** = Low.

U — User Readiness

The extent to which clinicians, administrators, and patients are prepared and willing to engage with the eHealth system.

Construct	High (H)	Medium (M)	Low–Medium (L–M)	Low (L)
Usability	Documented favourable usability scores; SUS/clinician-satisfaction surveys consistently ≥ 75	Mixed survey results; some pain but no widespread crisis	Clinician satisfaction below targets in published audits	Persistent, documented usability crisis (e.g., GAO findings of dissatisfaction)
Workflow integration	Designed with clinician co-design; embedded in existing workflows	Partial workflow embedding; some retrofit required	Workflow disruption acknowledged in audits	Documented workflow disruption with productivity loss
Training adequacy	Funded, multi-modal training programme deployed pre-go-live	Training exists but variable quality across sites	Training gaps identified in audits	No structured training programme
Trust / perceived usefulness	Clinician advocates speak publicly in favour of the system	Mixed clinician advocacy	Clinician resistance documented in formal surveys	Active clinician opposition or resistance
Patient engagement	Patient portal adoption $\geq 50\%$ nationally	Patient portal exists; adoption variable	Portal exists but low usage	No patient-facing functionality

Rating decision rule: Two or more constructs meeting H criteria with no constructs at L → **H**. Mix of M with one or two H/L → **M**. Two or more constructs at L–M with usability or training gaps documented in audits → **L–M**. Documented persistent user-satisfaction crisis (e.g., GAO findings) → **L**.

T — Technical Interoperability

The capacity of eHealth systems to exchange, interpret, and use clinical data across organisational boundaries.

Construct	High (H)	Medium (M)	Low–Medium (L–M)	Low (L)
Semantic interoperability	National terminology (SNOMED CT, LOINC); enforced	Standards adopted but not nationally bound	Partial standards adoption	No common terminology
Syntactic interoperability	HL7 FHIR R4 + CDA mandated; conformance testing	FHIR adoption variable across vendors	Legacy formats pre-dominant	No interoperability standard
System quality	Documented uptime ≥99.5%; mature tooling	Acceptable uptime; episodic incidents	Reliability concerns in audits	System reliability crisis
Data completeness	National data set obligation enforced	Voluntary upload with high participation	Variable coverage	Sparse coverage
API / exchange layer	National exchange backbone deployed (e.g., X-Road)	Federated exchange (HIE network) operational	Bilateral interfaces only	No exchange
Integration burden	Vendor agnostic; standards-driven integration	Some vendor lock-in but multiple options	Heavy custom integration required	Vendor monoculture / lock-in

Rating decision rule: National exchange backbone + enforced standards + uptime → **H**. Federated/standards-based but voluntary → **M**. Standards adopted but with substantial integration debt → **L–M**. No common standards / pervasive bilateral interfaces → **L**.

I — Institutional Alignment

The degree to which regulatory, policy, governance, and legal environments support and incentivise eHealth adoption.

Construct	High (H)	Medium (M)	Low–Medium (L–M)	Low (L)
Regulatory mandate	Statutory obligation to participate (e.g., national EHR law)	Strong incentive programme (HITECH-style) without mandate	Voluntary framework	policy No coordinating policy
National digital health strategy	Documented, funded, multi-year national strategy	Policy framework with episodic funding	Strategy in development	No national strategy
Information governance	Mature data-sharing agreements; national opt-out	Information governance frameworks exist; local variation	Information governance gaps acknowledged	No coherent governance
Privacy law alignment	Health-IT-specific privacy law (HIPAA, GDPR + national act)	Generic privacy law applied to health data	Privacy gaps acknowledged	No applicable framework
Interoperability mandate	Information blocking prohibited and enforced	Interoperability standards required (e.g., Cures Act)	Voluntary interoperability targets	No interoperability obligation
Incentive alignment	Direct financial incentives for adoption + use	Indirect incentives (quality reporting)	Limited incentive structure	No incentive structure

Rating decision rule: Statutory mandate + enforced governance + national strategy → **H**. Strong incentives + interoperability rules without mandate → **M** to **H** depending on enforcement. Voluntary frameworks with documented gaps → **L–M**. Absent or fragmented regulatory environment → **L**.

O — Organizational Execution

The internal organisational capabilities and leadership practices that determine whether an eHealth adoption strategy is planned, resourced, governed, and executed effectively.

Construct	High (H)	Medium (M)	Low–Medium (L–M)	Low (L)
Leadership commitment	Sustained executive sponsorship; named accountable owners	Leadership engagement at launch; sustainability variable	Leadership turnover or disengagement during execution	No accountable owner
Implementation planning	Documented multi-phase plan with milestones; published	Plan exists; milestones missed without remediation	Reactive rather than planned execution	No formal plan
Change management	Funded change-management capability throughout lifecycle	Change-management activities at go-live only	Change management identified as gap in audits	No change-management capacity
Stakeholder engagement	Clinician co-design + ongoing feedback channels	Stakeholder engagement at launch; episodic after	Stakeholder voice limited in execution	No engagement mechanism
Vendor governance	Mature vendor-management practice; contract performance tracked	Contract management functional	Vendor disputes documented in audits	Vendor relationship dysfunctional
Resource capacity	IT staff, financial, and clinical-time resources adequate	Resource constraints acknowledged but managed	Resource gaps cited as adoption barrier	Severe resource constraint
Issue management	Proactive issue identification and resolution	Reactive issue resolution	Issue backlog documented in audits	No issue-management process

Rating decision rule: All seven constructs meeting at least M, with leadership and change-management at H → **H**. Mixed M/L–M with vendor or change-management weakness → **M**. Two or more constructs at L–M, audit findings of execution gaps → **L–M**. Documented severe execution failures (cost overruns, schedule slips, audit findings of inadequate management) → **L**.

Evidence-Anchored Ratings for the Four Cases

For each case, ratings are anchored to specific sources from the bibliography (citation keys in parentheses). This is the audit trail a reviewer or replicator can use to challenge or reproduce any rating.

Case 1: U.S. ONC HIE — U(M) T(H) I(H) O(M)

Domain	Rating	Evidence anchor
U	M	[2] documents variable clinician engagement and adoption-rate gaps across participating entities; [9] identifies clinician trust and workflow concerns as persistent HIE barriers; [10] lists usability and training as adoption inhibitors.
T	H	[2] references FHIR API mandates under the Cures Act; [14] describes the U.S. as one of five mature HIE-policy nations; nationwide TEFCA infrastructure deployed.
I	H	HITECH Act incentive program drove U.S. EHR adoption from ~10% to >80% ([2]); 21st Century Cures Act information-blocking rules now enforced ([14]); strong statutory framework.
O	M	[2] documents variable participation rates across HIEs; organisational uptake heterogeneity is the primary O-domain weakness; [15] describes execution as the differentiator across national programs.

Case 2: NHS Shared Care Records (UK) — U(M) T(H) I(H) O(M)

Domain	Rating	Evidence anchor
U	M	[20] acknowledges variable local clinician engagement; sustained training and workflow integration cited as central challenges.
T	H	[20] describes national interoperability standards plus a shared-care-record platform; cross-organisational information sharing operational.
I	H	NHS Long Term Plan + national data opt-out + information governance frameworks documented in [20]; central national strategy authority.
O	M	[20] flags local information-governance variability and workflow integration as persistent execution challenges; [15] lists O-domain factors as decisive across national EHR rollouts.

Case 3: Estonia eHealth — U(H) T(H) I(H) O(H)

Domain	Rating	Evidence anchor
U	H	[3] reports near-comprehensive coverage and high clinician engagement; 99% patient e-Health Record adoption widely cited.
T	H	[3] describes the X-Road secure data-exchange layer; [14] cites Estonia as architectural reference; enforced national standards.
I	H	[28] documents the legal mandate for provider participation; 2008 Estonian Health Information System Act provides statutory backbone.
O	H	[3] describes sustained ministry-level governance and evolving operational frameworks; consistent execution across two decades of national infrastructure operation.

Case 4: U.S. VA/DoD EHR Modernization — U(L–M) T(L–M) I(H) O(L–M)

Domain	Rating	Evidence anchor
U	L–M	[5] documents persistent clinician dissatisfaction at deployed VA sites; [6] cites user-satisfaction scores below targets at DoD sites. The rubric’s U decision rule lists “Documented persistent user-satisfaction crisis (e.g., GAO findings) → L”; the L–M rating reflects the documented crisis tempered by ongoing remediation activity at deployed sites. <i>Revised from M (2026-04-30); see audit-trail note below.</i>
T	L–M	[5] describes mature commercial Oracle Health platform but substantial configuration burden integrating with legacy VistA and DoD clinical systems; documented system-quality and data-completeness gaps at deployed sites. The rubric’s T decision rule “Standards adopted but with substantial integration debt → L–M” is the closest match. <i>Revised from M (2026-04-30); see audit-trail note below.</i>
I	H	Congressional mandate via FY2018 NDAA; sustained appropriations; clear executive-branch authority — [5], [6] both confirm institutional backing.
O	L–M	[5] flags inadequate vendor governance and slow remediation; [6] cites issue-management process inadequacies and schedule slips. Persistent execution gaps drive the lowest-tier ratings in the four-case set.

Audit-trail note (2026-04-30): The U and T ratings for VA/DoD were revised from M to L–M following the AI-assisted second-rater pass documented in Supplementary File S4. The second rater applied the rubric’s explicit decision-rule trigger language for both domains; the original ratings (M for both) were a band more lenient than the rubric strictly indicated. Revised ratings retain a middle position (L–M rather than the second rater’s L for U) to reflect ongoing remediation activity at deployed sites. Inter-rater agreement statistics in Supplementary File S4 are computed on the *pre-revision* ratings, as is methodologically appropriate.

How to reproduce: A second rater should retrieve the cited source(s) for each cell, apply the rubric criteria above, and assign a rating. Rating disagreements should be reported as an inter-rater reliability outcome.

Limitations of this Rubric

1. **Single-rater application.** Ratings in the current manuscript were assigned by one reviewer (the author). Independent dual-rating with inter-rater reliability is not yet performed.
2. **Categorical resolution.** Four bands (H / M / L–M / L) are coarse. A reviewer wanting finer resolution could move to a 5- or 7-point ordinal scale.
3. **Construct weighting.** Each domain treats its constructs as equally informative; in practice, some constructs (e.g., regulatory mandate within I) may dominate the rating. This is not formalised here.

4. **Evidence weighting.** Government audit findings and peer-reviewed literature are weighted equally; some reviewers might down-weight grey literature.

Last updated: 2026-04-29

Supplementary File S4: Second-Rater Protocol — U-T-I-O Case Ratings

Purpose: Mitigate the single-rater limitation on the case-level U-T-I-O ratings ($4 \text{ cases} \times 4 \text{ domains} = 16$ ordinal ratings on a four-band scale: H / M / L-M / L). Distinct from the 17-source domain-presence self-audit (audit-check.md), which audits *domain-presence* coding (Y/blank) on the 17-source corpus.

Scope: Applies the rubric in Supplementary File S3 to the four cases (U.S. ONC HIE, NHS Shared Care Records, Estonia eHealth, U.S. VA/DoD EHR Modernization). Each case-domain cell is re-rated by a second rater blind to the original ratings, then compared and adjudicated.

1. Method

1.1 Rater Independence

The second rater receives: (a) the construct-level criteria for U, T, I, and O (the four rubric tables in Supplementary File S3); (b) the case-by-case evidence anchors with citation keys, but **with the original ratings redacted**; and (c) the bibliography for source verification. The second rater does **not** see the original H / M / L-M / L assignments before producing their own.

1.2 Rater Types Reported

This protocol supports two rater types, reported separately:

1. **Human second rater** (preferred; not yet recruited). A credentialed reviewer with health-IT or implementation-science expertise re-rates from the blinded rubric. Status: pending.
2. **AI-assisted second rater** (supplementary; executed). Anthropic Claude Opus 4.7, accessed 2026-04-29. Given the blinded rubric and access to the cited primary sources via web tools. **Does not substitute** for human dual-rating; mirrors the AI-assisted pattern used in the 17-source domain-presence self-audit.

1.3 Agreement Statistics

With $N = 16$ ordinal cells on a 4-band scale, three statistics are reported jointly:

- **Percent exact agreement** — fraction of cells where second rater matches original exactly (primary; easy to interpret).
- **Percent within-one-band agreement** — fraction where ratings are within one band of each other (catches close disagreements that may reflect rubric-band ambiguity).

- **Linear-weighted Cohen’s κ** — ordinal-aware agreement statistic. Weights: $w_{ij} = |i - j|/3$ where i, j are band indices on a 1–4 scale (L=1, L–M=2, M=3, H=4). Reported with the explicit caveat that **N = 16 yields a high-variance κ point estimate**; reported for transparency, not as a robust inferential claim.

1.4 Disagreement Adjudication

For each disagreement (exact mismatch), the original rater records: which construct(s) drove the original rating; which construct(s) the second rater appears to have weighted differently; and whether the disagreement reflects (a) different evidence interpretation, (b) rubric-band ambiguity, or (c) a defensible rating revision. If category (c), the original rating is updated in Supplementary File S3 with a logged change.

2. Blinded Rubric Handed to the Second Rater

Note for second rater: Use the construct definitions in Supplementary File S3 and the rating decision rules. For each case below, read the cited sources and assign one of {H, M, L–M, L} per domain. Do not consult the evidence-anchored ratings section of Supplementary File S3 before completing your ratings.

Case A — U.S. ONC HIE. Primary sources: [2], [14], [9], [10]. Context: Federated, market-based HIE infrastructure under HITECH Act incentives and 21st Century Cures Act / TEFCA. Rate U / T / I / O.

Case B — NHS Shared Care Records (UK). Primary sources: [20], [15]. Context: Centrally coordinated national interoperability programme within a single-payer health system. Rate U / T / I / O.

Case C — Estonia eHealth. Primary sources: [3], [28], [14]. Context: Legally mandated national digital health infrastructure on the X-Road exchange platform, operational since 2008. Rate U / T / I / O.

Case D — U.S. VA/DoD Federal EHR Modernization. Primary sources: [5], [6]. Context: Large-scale federal EHR replacement (Oracle Health / Cerner; MHS GENESIS) under FY2018 NDAA mandate. Rate U / T / I / O.

Blank rating sheet:

Case	U	T	I	O
A — U.S. ONC HIE				
B — NHS Shared Care Records				
C — Estonia eHealth				
D — VA/DoD EHR				

3. Results

3.1 AI-Assisted Second Rater (Anthropic Claude Opus 4.7, 2026-04-30)

The AI rater was given the rubric criteria (§§U–O of Supplementary File S3) and the blinded case definitions in §2 above. The rater operated in a fresh agent context with no exposure to the original ratings. **Audit-trail note:** A first AI pass was discarded after the rater reported inadvertent exposure to original ratings in an earlier draft. The ratings below are from the clean second pass; the discarded pass is not retained.

Table ST3: AI-Assisted Second-Rater Ratings and Justifications

Case	Dom	Rating	Justification (≤ 2 sentences; constructs cited)
A — U.S. ONC U HIE		L–M	Usability and workflow integration mixed across the federated HIE landscape with documented clinician burden ([9], [10]); Patient engagement via portals exists but adoption is variable and Training adequacy varies widely across participating organisations.
A — U.S. ONC T HIE		M	Syntactic interoperability advancing under FHIR/USCDI and the API/exchange layer is federated (TEFCA QHINs, eHealth Exchange) and operational, but Semantic interoperability and Data completeness remain voluntary and uneven — fits “federated/standards-based but voluntary $\rightarrow$ M”.
A — U.S. ONC I HIE		M	Strong Interoperability mandate (Cures Act information-blocking) and Incentive alignment (HITECH/MIPS) plus HIPAA Privacy law alignment are in place, but no unified statutory mandate to participate in HIE and Information governance varies state-by-state.
A — U.S. ONC O HIE		M	Leadership commitment at federal ONC level is sustained and Implementation planning documented across multi-year reports to Congress, but Stakeholder engagement and Change management at the provider level are uneven.
B — NHS U Shared Care		M	Workflow integration and Trust improving as ShCRs embed into clinician workflows across ICSs, but Usability scores are mixed across regional implementations and Training adequacy varies across the 42 ICS footprints.

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Table ST3: AI-Assisted Second-Rater Ratings (continued)

Case	Dom	Rating	Justification (≤ 2 sentences; constructs cited)
B — NHS T Shared Care		M	Syntactic interoperability uses FHIR UK Core and Semantic interoperability leverages SNOMED CT, with a federated regional API/exchange layer operational, but Data completeness and System quality vary across ICS-level instances rather than a single national backbone.
B — NHS I Shared Care		H	National digital health strategy is documented and funded (NHS Long Term Plan, ShCR programme), Privacy law alignment is strong (UK GDPR + Data Protection Act + Caldicott), and Information governance is mature with national IG frameworks.
B — NHS O Shared Care		M	Leadership commitment at NHS England and Implementation planning are documented, but Stakeholder engagement and Change management vary by ICS and Resource capacity is acknowledged as constrained ([15]).
C — Estonia U		M	Patient engagement is genuinely high (portal usage well above 50%) and Trust among clinicians is generally favourable, but published Usability scores and Training adequacy detail are limited in the cited sources, preventing a confident H across all five constructs.
C — Estonia T		H	National X-Road API/exchange layer is the canonical example of a national exchange backbone, with enforced Semantic and Syntactic interoperability standards, mandated Data completeness via the national EHR obligation, and documented high System quality — meets the H decision rule cleanly.
C — Estonia I		H	Statutory Regulatory mandate (national EHR law since 2008) plus a documented multi-year national digital health strategy, mature Information governance with patient opt-out, GDPR-aligned Privacy law, and an enforced Interoperability mandate — satisfies “statutory mandate + enforced governance + national strategy $\rightarrow$ H”.

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Table ST3: AI-Assisted Second-Rater Ratings (continued)

Case	Dom	Rating	Justification (≤ 2 sentences; constructs cited)
C — Estonia	O	M	Leadership commitment and Implementation planning sustained over 15+ years with stable Vendor governance and Resource capacity, but cited sources provide limited evidence on Change management and Issue management practices at the operator level.
D — VA/DoD	U	L	GAO findings document a persistent Usability crisis (clinician dissatisfaction at deployed VA sites), Workflow disruption with productivity loss, and Training adequacy gaps — meets the explicit “Documented persistent user-satisfaction crisis (e.g., GAO findings)” trigger for L.
D — VA/DoD	T	L–M	Oracle Health/Cerner and MHS GENESIS use HL7/FHIR standards (not “no common standards”), but System quality issues (outages, patient-safety incidents) and Data completeness / Integration burden problems documented in GAO reports indicate substantial integration debt → L–M.
D — VA/DoD	I	M	Statutory direction via FY2018 NDAA and federal Privacy law alignment (HIPAA + federal records statutes) provide strong policy scaffolding, but Information governance and Interoperability mandate enforcement between VA, DoD, and community providers remain partially realised per GAO — fitting strong-incentives-without-full-mandate M.
D — VA/DoD	O	L	GAO documents severe execution failures — cost overruns, schedule slips, deployment pauses, Vendor governance disputes with Oracle, and audit findings of inadequate Change management, Issue management, and Resource capacity — meets the L decision rule for “documented severe execution failures”.

3.2 Comparison with Original Ratings

Case	Dom.	Original	AI Rater	Exact	Within-one
A — U.S. ONC HIE	U	M	L–M	No	Yes
A — U.S. ONC HIE	T	H	M	No	Yes
A — U.S. ONC HIE	I	H	M	No	Yes
A — U.S. ONC HIE	O	M	M	Yes	Yes
B — NHS Shared Care	U	M	M	Yes	Yes
B — NHS Shared Care	T	H	M	No	Yes
B — NHS Shared Care	I	H	H	Yes	Yes
B — NHS Shared Care	O	M	M	Yes	Yes
C — Estonia	U	H	M	No	Yes
C — Estonia	T	H	H	Yes	Yes
C — Estonia	I	H	H	Yes	Yes
C — Estonia	O	H	M	No	Yes
D — VA/DoD	U	M	L	No	No
D — VA/DoD	T	M	L–M	No	Yes
D — VA/DoD	I	H	M	No	Yes
D — VA/DoD	O	L–M	L	No	Yes

Note: Original ratings are pre-revision values, as methodologically appropriate for inter-rater reliability computation. Two cells (D-U and D-T) were subsequently revised; see §4.

3.3 AI-Pass Agreement Tally

Statistic	Value
N	16
Exact agreement	6/16 (37.5%)
Within-one-band agreement	15/16 (93.8%)
Linear-weighted Cohen’s κ	0.27 (Landis–Koch: “fair”)
Marginal distribution — original	L=0, L–M=1, M=6, H=9
Marginal distribution — AI rater	L=2, L–M=2, M=9, H=3
Direction of disagreements	All 10: AI more conservative (lower band)
Cells >1 band apart	1/16 (D — VA/DoD U: M vs. L)

Reproducibility note: κ computed with linear weights $w_{ij} = |i - j|/3$ over the four-band ordinal scale (L=1, L–M=2, M=3, H=4). At N=16 the κ point estimate is high-variance; the within-one-band percentage is the more interpretable agreement signal.

Interpretation: The AI rater was systematically more conservative than the original across all 10 disagreements. The marginal-distribution shift—original modal H, AI modal M—depresses κ even though within-one-band agreement is 94%. Two cells (D–U, D–T) are flagged in §4 as candidates for revision because the rubric’s explicit decision-rule trigger language more closely matches the AI’s rating.

3.4 Human Second Rater

Status: pending recruitment. The same blinded rubric (§2) and rating sheet apply.

4. Adjudication Log

For each of the 10 exact mismatches, the disagreement is classified as **(a) different evidence interpretation**, **(b) rubric-band ambiguity**, or **(c) defensible rating revision**. Disposition: **kept** (original retained); **revised** (original updated); **flagged** (unchanged pending external second-rater confirmation).

Table ST4: Adjudication Log for All 10 Exact Mismatches

Cell	Orig $\rightarrow$	Δ	Class	Reasoning	Disposition
A-U	M $\rightarrow$ L- M	1	(b) Rubric ambiguity	M/L-M boundary is the fuzziest band on the U domain. Original M cites variable adoption + workflow concerns. AI L-M cites the same evidence and applies the “two or more constructs at L-M with usability or training gaps documented in audits” decision rule. Both are defensible.	Kept
A-T	H $\rightarrow$ M	1	(a) Evidence interp.	Original weights Cures Act + FHIR API mandate + TEFCA infrastructure as H. AI weights TEFCA’s federated/voluntary structure + non-bound semantic standards as fitting “Federated/standards-based but voluntary $\rightarrow$ M”.	Flagged — borderline H/M; Cures Act enforcement is a defensible H-tipping factor
A-I	H $\rightarrow$ M	1	(a) Evidence interp.	Original weights Cures Act enforcement + HITECH + HIPAA as collectively H per the rubric’s “enforcement clause.” AI treats absence of statutory participation mandate + state-by-state IG variation as M.	Kept — H is consistent with rubric’s enforcement clause
B-T	H $\rightarrow$ M	1	(a) Evidence interp.	Original weights national interoperability standards + ShCR platform as H. AI cites federated regional implementation across 42 ICSs and variable Data completeness as M.	Flagged — borderline H/M; federated regional structure is a genuine H constraint
C-U	H $\rightarrow$ M	1	(a) Evidence interp.	Original weights $\sim 99\%$ patient e-Health Record adoption + clinician advocacy as H. AI cites limited published Usability/Training data in cited sources as preventing a five-construct H.	Kept — patient engagement and trust evidence supports H per U decision rule

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Table ST4: Adjudication Log (continued)

Cell	Orig →	Δ	Class	Reasoning	Disposition
C-O	H → M	1	(a) Evi- dence in- terp.	Original weights sustained ministry-level gov- ernance over 15+ years as H. AI cites lim- ited evidence on Change management and Is- sue management at operator level in cited sources.	Flagged — H is supported by execution longevity but construct- level evidence partial
D-U	M → L	2	(c) Revised	Rubric U decision rule states: “Documented persistent user-satisfaction crisis (e.g., GAO findings) → L.” [5] and [6] are GAO find- ings of persistent dissatisfaction. Original M was more lenient than the rubric strictly in- dicates.	Revised: M → L–M (2026-04-30). Middle ground reflecting doc- umented crisis plus ongoing remediation; not pure L.
D-T	M → L– M	1	(c) Revised	Rubric T decision rule: “Standards adopted but with substantial integration debt → L– M.” GAO documents Oracle/Cerner integra- tion debt, deployment pauses, system quality issues. AI’s L–M tracks the rubric language more directly than original M.	Revised: M → L–M (2026-04-30). Aligns with rubric trigger language.
D-I	H → M	1	(a) Evi- dence in- terp.	Original weights FY2018 NDAA mandate + congressional appropriations as H. AI cites partial mandate-enforcement between VA, DoD, and community providers as M.	Kept — statu- tory mandate + sustained appropriations support H

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Table ST4: Adjudication Log (continued)

Cell	Orig →	Δ	Class	Reasoning	Disposition
D-O	L– M → L	1	(b) Rubric ambigu- ity	L/L–M boundary on O is fuzzy. Original L– M acknowledges audit findings of execution gaps. AI L applies “documented severe ex- ecution failures” trigger — GAO reports do document cost overruns, schedule slips, and inadequate management.	Kept — but AI’s L is de- fensible; a third rater confirming would tighten the case

Disposition	Count
Kept (original retained)	5
Flagged — left for external second rater	3
Revised (rubric trigger language matched by AI rater)	2
Total disagreements	10

Revisions applied (2026-04-30): Two cells in the VA/DoD case were revised from M to L–M. The VA/DoD case row was updated from U(M) T(M) I(H) O(L–M) to U(L–M) T(L–M) I(H) O(L–M) in the rubric (Supplementary File S3, Case 4) and propagated to the manuscript results table and conclusion.

Important methodological note: The agreement statistics in §3.3 (6/16 exact, 15/16 within-one-band, $\kappa = 0.27$) are computed on the *pre-revision* ratings. Recomputing post-revision would be circular and is not reported.

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